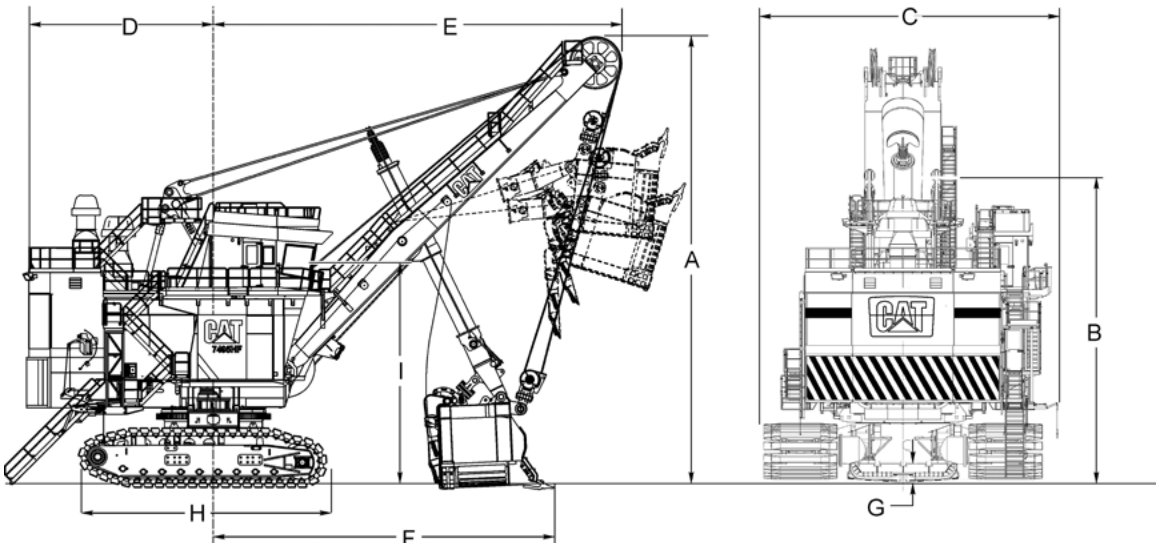
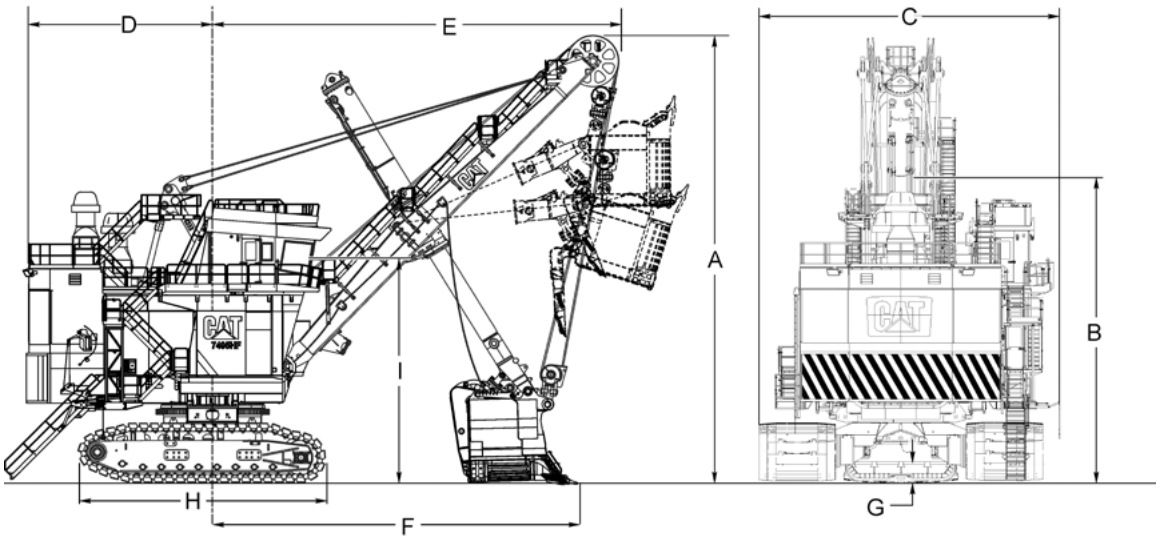


MODEL		7495 HF	
Boom Length		20.40 m	**67'0"**
A	Height	20.87 m	**68'6"**
B	A-Frame Height	14.00 m	**46'0"**
C	Overall Width	13.96 m	**46'1"**
D	Tail Swing Radius	9.30 m	**30'8"**
E	Clearance Radius	19.65 m	**64'5"**
F	Radius of Level Floor*	17.00 m	**55'11"**
G	Minimum Ground Clearance	0.90 m	**3'0"**
H	Track Length	11.43 m	**37'6"**
I	Operator Eye Level	10.61 m	**34'10"**

*Dimensions based on 45 m^3 **(59 yd^3)** dipper.
Information subject to change.
All dimensions are approximate and will vary depending on dipper size.



MODEL		7495 HF HydraCrowd	
Boom Length		20.40 m	**67'0"**
A	Height	20.87 m	**68'6"**
B	A-Frame Height	14.00 m	**46'0"**
C	Overall Width	13.96 m	**46'1"**
D	Tail Swing Radius	9.30 m	**30'8"**
E	Clearance Radius	19.65 m	**64'5"**
F	Radius of Level Floor*	17.00 m	**55'11"**
G	Minimum Ground Clearance	0.90 m	**3'0"**
H	Track Length	11.43 m	**37'6"**
I	Operator Eye Level	10.61 m	**34'10"**

*Dimensions based on 45 m^3 **(59 yd^3)** dipper.
Information subject to change.
All dimensions are approximate and will vary depending on dipper size.

GROUND BEARING PRESSURE

Ground bearing pressure, or the amount of weight that can be supported by ground conditions, determines the link width (and occasionally the machine model) that can be used at a mine site. Softer ground conditions require wider links while the softest conditions (i.e. oil sands) require a specific model (7495 HF) which was designed to operate under low ground bearing pressure conditions. A link too narrow for the application may even cause the crawlers to sink into the ground.

The chart below lists the link length options for each electric rope shovel model as well as the corresponding minimum ground bearing pressures.

MODEL	7295[1]		7395[2]		7495 HD[3]		7495[4]	
Standard Link Length	183 cm	**72"**	213 cm	**84"**	213 cm	**84"**	200 cm	**79"**
Ground Bearing Pressure	269 kPa	**39 psi**	371 kPa	**54 psi**	400 kPa	**58 psi**	380 kPa	**55 psi**
Optional Link Length	—		274 cm	**108"**	274 cm	**108"**	259 cm	**102"**
Ground Bearing Pressure	—		276 kPa	**43 psi**	316 kPa	**46 psi**	297 kPa	**43 psi**

MODEL	7495 HydraCrowd[4]		7495 HF[5]		7495 HF HydraCrowd[5]	
Standard Link Length	200 cm	**79"**	318 cm	**125"**	318 cm	**125"**
Ground Bearing Pressure	382 kPa	**55 psi**	247 kPa	**36 psi**	248 kPa	**36 psi**
Optional Link Length	259 cm	**102"**	—		—	
Ground Bearing Pressure	298 kPa	**43 psi**	—		—	

[1] Based on 19 m³ **(25 yd³)** dipper.
[2] Based on 24 m³ **(32 yd³)** dipper.
[3] Based on 32 m³ **(42 yd³)** dipper.
[4] Based on 56 m³ **(73 yd³)** dipper.
[5] Based on 45 m³ **(59 yd³)** dipper.
Information subject to change.
All dimensions are approximate.

DIPPER SELECTION

Selecting the correct dipper size plays an integral role in maximizing productivity from a truck shovel operation. Optimal dipper size is a function of truck size, material loose density, and fill factor.

The first step in calculating dipper size is to determine the machine's optimal payload based on truck size. To calculate this, divide truck payload by three and four (three or four pass even loading is ideal for high productivity). Compare these values to the machine's maximum payload. The machine's optimal payload is the largest value that is less than the maximum payload.

For example, consider a mine operating a 7495 shovel (maximum payload = 120 tons) and 797 trucks (payload = 400 tons). 400 ÷ 3 = 133.3 and 400 ÷ 4 = 100. 133.3 can be eliminated as it is greater than the shovel's maximum payload (120 tons). 100 tons is the largest value less than the machine's maximum payload, and therefore, is the machine's optimal payload.

Once optimal machine payload is determined, material loose density and fill factor are used to calculate the optimum dipper size. To calculate material loose density, divide insitu (undisturbed material density) by 1 + swell factor.

$$\text{Material loose density} = \frac{\text{Insitu}}{(1 + \text{swell factor})}$$

Once material loose density is calculated, use the equation below to find the optimal dipper size. Fill factor is affected by the dipper geometry, rake angle, and material properties, including fragmentation and critical angle of repose. Fill factors are generally between 90% and 115% for efficiently-sized dippers.

$$\text{Optimal dipper size} = \frac{\text{Optimal machine payload}}{\text{Material loose density}} \times \text{Fill factor}$$

For example: For a mine with the following conditions:

Insitu: 2.276 ton/yd³
Swell factor: 35%
Fill factor: 95%
Optimal machine payload: 100 tons

$$\text{Optimal dipper size} = \frac{100 \text{ tons}}{2.276 \div (1 + 0.35)} \times .95 = 62.5 \text{ yd}^3$$

TRUCK MATCH

Truck shovel match, or the number of passes necessary to load a truck, has a major impact on an operation's productivity. To optimize productivity, 3 to 4 even pass loading of trucks is ideal. The chart below shows pass match between Cat electric rope shovels and Cat trucks.

	Payload	7295	7395	7495 HD	7495	7495 HF
MODEL	tonne **(ton)**	45 **(50)**	64 **(70)**	82 **(90)**	109* **(120*)**	109* **(120*)**
785D	136 **(150)**	**3**				
789D	181 **(200)**	**4**	**3**			
793F	227 **(250)**		**4**	**3**		
794 AC	291 **(320)**			**4**	**3**	**3**
795F AC	313 **(345)**			**4**	**3**	**3**
797F	363 **(400)**				**4**	**4**
796 AC	327 **(360)**			**4**	**3**	**3**
798 AC	372 **(410)**				**4**	**4**

*Indicates maximum payload.

RATED SUSPENDED LOAD

Rated Suspended Load is the designed maximum load limit for any Cat Electric Rope Shovel (ERS) model/machine serial number. The maximum load limit is defined as the combined total of Dipper Weight plus Struck Payload Weight.

- **Dipper Weight:** Dipper Weight is the combined total dead weights of dipper body, GET, lip and corner shrouds, dipper door, door latch assembly, wear liner packages for body and door, snubbers, pitch braces, padlocks and pins/pin retainers. (**Note:** Handle weight is NOT included in calculation of Dipper Weight.)
- **Struck Payload Weight:** Struck Payload Weight is the live weight of the material in the dipper. This is a calculated payload weight based on the struck volume of the dipper multiplied by the density of the loose material in the dipper.

  Struck volume of a dipper is the contained interior volume of the dipper from the closed door up to the plane of a straight line from the front edge of the dipper back (top) to a point on the dipper lip (floor) where the teeth are attached.

The below values for Rated Suspended Loads (RSL) for Cat Electric Rope Shovels are applicable to shovels commissioned on or after June 1, 2016. Since Cat Electric Rope Shovels have seen numerous upgrades throughout their history which will affect shovel capability, a Rated Suspended Load (RSL) value based on the specific configuration of each machine is more appropriate for machines manufactured or commissioned prior to that date.

ERS MODEL	Rated Suspended Load (RSL)	
7295	81 647 kg	**180,000 lb**
7395	117 934 kg	**260,000 lb**
7495 HD	154 221 kg	**340,000 lb**
7495 with 100 tonnes **(110 short tons)** Payload	185 973 kg	**410,000 lb**
7495 HF with 100 tonnes **(110 short tons)** Payload	185 973 kg	**410,000 lb**
7495 with 109 tonnes **(120 short tons)** Payload	195 045 kg	**430,000 lb**
7495 HF with 109 tonnes **(120 short tons)** Payload	195 045 kg	**430,000 lb**

10/10/20 LOAD MANAGEMENT POLICY — ELECTRIC ROPE SHOVELS

"Machine overload" shall consist of operating the shovel(s) outside of the acceptable allowable load distribution.

The customer shall be responsible for the monitoring and management of all shovel operational activities, including, but not limited to, management of the machine loading such that its loaded distribution is within the acceptable allowable load distribution.

Operating outside of these parameters over any rolling 30 day period can void the customer's standard warranty; any extended warranty; any availability guarantee and/or result in adjustments to Caterpillar's obligations under any contract or agreement.

Loading Distribution

Loading use over any rolling 30 Day period analyzed:

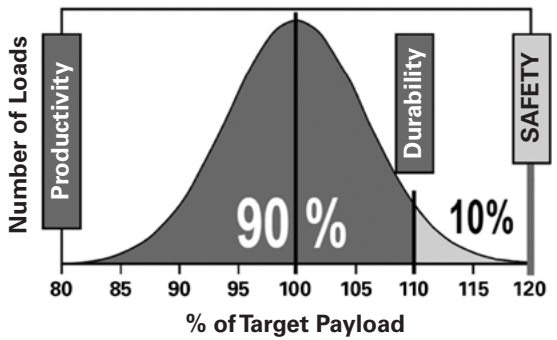
- No more than 10% of loads should exceed 110% of rated payload.
- No load should exceed 120% of rated payload.
- The average of all loads for the unit not to exceed the rated payload.

KEY

90% of loads should fall into this range.

No more than 10% of loads should exceed 110% of the target payload.

No loads should be above 120% of the target payload.



Loading use in excess of this prescribed distribution is considered "machine overload."

Notes

- For this policy to be in effect, the "operational cycle" payload feedback must be available to the operator in order that the customer can control actual payloads.
- Customer must procure, install, and use Caterpillar payload measurement and monitoring systems.
- Caterpillar must be notified in a timely fashion if the payload monitoring and feedback system is out of service.

HYDRAULIC MINING SHOVELS

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HYDRAULIC MINING SHOVELS

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Cycle Times 13-22
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GENERAL INTRODUCTION

Hydraulic Mining Shovels

Hydraulic mining shovels are designed to move large volumes of materials, and the majority are employed in the mining industry for loading large Mining and Off-Highway Trucks. Key design characteristics are high digging forces and quick working cycles for best performance, and a durable design to handle severe working conditions. Hydraulic mining shovels dig and load various commodities, such as gold, copper, iron ore, diamonds and coal, or they remove overburden, in any climatic condition around the world. Typically operated around the clock, Hydraulic Mining Shovels accumulate up to 6500 hours per year.

Nomenclature

The 6000 Series constitutes the Cat hydraulic mining shovel product line. The name of each individual model gives a direct reference to the size of the machine and consists of a four-digit number.

The first digit designates the 6000 Series and is followed by another three digits indicating the approximate bucket payload of that model in metric tonnes. 6015 and 6020B models are only available with a backhoe attachment and buckets while 6030 and larger models are also available with a face shovel attachment.

The following table gives a summary of all versions currently available.

Backhoe	Front Shovel
6015	N/A
6020B	N/A
6030	6030
6040	6040
6060	6060

N/A = Not Available

Technical Specifications

All data given herewith is subject to change. The specalogs/spec sheets for individual models may have been updated after publishing of this Performance Handbook edition.

Contact Details

If there are any questions or comments regarding the information on hydraulic mining shovels given in this chapter please contact:

Feddern_Nils@cat.com or
Tegtmeier_Dirk@cat.com

TriPower

On all face shovel (FS) models, Caterpillar uses a unique boom design, called TriPower, that employs rotatable triangular rockers to generate superior mechanical leverage and control. This unique Caterpillar design results in:

- Increased effective lifting force
- Constant boom momentum
- Automatic constant bucket angle position in horizontal and vertical direction
- Automatic roll-back limiter

Furthermore, the TriPower system enables Cat hydraulic front shovels to use smaller-diameter boom cylinders. This benefit results in faster lifting speeds.

Superior Oil Cooling System

From 6020B up to 6060 the independent oil cooling system utilizes dedicated pumps to provide cooling capacity as needed, whether the engine is idling or under load, whereas conventional cooling systems only provide cooling when the engine is under load and the machine is working. Consequently, on these models the hydraulic oil circulates through the cooling system even when the machine is waiting for the next truck. This system provides a more efficient means of cooling, particularly in demanding applications. The radiator fan speed is thermostatically controlled for greater efficiency.

Closed-Loop Swing Circuit

The Cat closed-loop swing circuit results in fast implement motions during swing. The kinetic energy of the swing motion is used during deceleration to support driving the main and auxiliary pumps. Compared to conventional open circuit systems, the Cat closed-loop swing circuit is more energy-efficient, generates less heat and delivers faster cycle times.

The 6020B includes an open-loop swing circuit due to its newly designed advanced hydraulics. Utilizing dedicated pump flow allocation technology, hydraulic pumps are allocated to individual circuits on demand, allowing all produced oil flow to be translated into cylinder motion, minimizing hydraulic inefficiencies and heat generation. To fully leverage the advantages of this innovative technology, the open-loop swing system was incorporated for its use of a common set of pumps to serve all functions and the flexibility it offers for utilization of hydraulic flow.

Simple and Efficient Hydraulic System

From 6020B up to 6060 the main hydraulic valve block is located on top of the boom. This design reduces the total number of hoses that are needed and ensures they are neatly organized for safe operation, easy inspection and fast service. Because float valves are used to lower the boom instead of engaging pumps, the boom moves faster and other operating functions can occur simultaneously, such as bucket curl and stick in/out. This results in faster cycle times.

Monitoring and Diagnostic System

Enhancing diagnostic capabilities and providing detailed troubleshooting functions, the machine control system uses sensors throughout the machine to monitor operating data, record faults, and notify the operator audibly and visually. This promotes the earliest possible detection of faults and allows for timely maintenance planning and assistance for speedy repair.

Machine Accessibility

Systems throughout Cat hydraulic mining shovels are designed for easy access, enhancing serviceability.

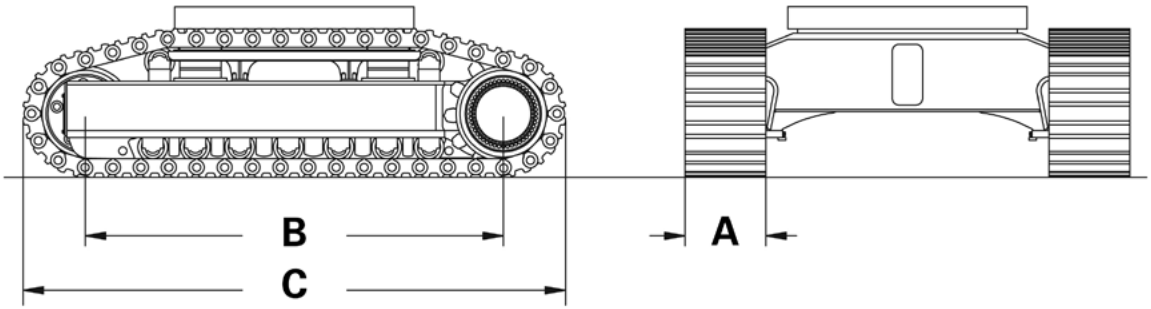
- **Hydraulic Valve Block** — The valve block is located on the boom where it's cleanly laid out and easily accessed by walkways on both sides. This reduces the number of hoses leading from the superstructure.
- **Superstructure** — Exceptional accessibility is provided to systems like the swing motor, swing gearbox and rotary distributor in the well-organized superstructure. The engine is accessible from three sides on most models.

GROUND PRESSURE

The ground pressure specifies the specific weight load transferred to the surface, and can be an indicator for the stability against subsidence. Though such conditions typically apply to smaller excavators in heavy earthmoving operations, there could be rare instances where this may apply in mining applications. Nevertheless, the vast majority of hydraulic mining shovels are delivered with standard track shoes.

The value is determined by setting the machine weight force in relation to the ground contact area of the tracks.

Ground Contact Area



Keys

A = Width of standard track shoe
B = Track length, idler center to sprocket center
C = Overall track length, grouser bar to grouser bar

To define the ground contact area on soft ground the following formula is used (ISO 16754):

$$2 \times A \times (B + (C - B) \times 0.35)$$

Optional Track Pads

In addition to the standard track shoes, optional track pads with different widths are available; however, these optional track pads are only procured on machine orders, and may therefore effect the lead time of the entire hydraulic mining shovel. Furthermore, a front shovel (FS) configured machine equipped with wider track pads increases the risk of track pad damage from bucket strikes.

BUCKET SELECTION

To maximize the durability and life expectancy of Hydraulic Mining Shovels, it is important to choose the correct bucket size based on the maximum loose density of the material to be loaded. Different bucket volumes for common material densities are available for most models. A loose density of 1.8 t/m³ (3030 lb/yd³) is considered standard and the respective bucket volume is suitable for the majority of applications. Additional bucket sizes are offered for lighter or heavier materials.

Rated Bucket Capacity

The following standards are used to determine the rated bucket capacity:

ISO 7451 for backhoe buckets
ISO 7546 for face shovel buckets

Rated bucket capacity for Hydraulic Mining Shovels is calculated with different bucket tilt angles for backhoe and face shovel buckets. The linkage geometry on backhoe attachments allows for larger bucket tilt angles allowing steaper heaps compared to face shovel buckets. Therefore, the nominal bucket capacity is calculated on backhoe buckets with a heap of 1:1, and on face shovel buckets with a heap of 2:1 as shown in the adjacent example sketch.

Nominal Bucket Material Density Ratings and Classification

Each bucket is classified using a nominal material density value that allows 100% fill factor using a standard configuration e.g. penetration style tips.

The following density categories are used:

Light Rock	Nominal loose density < 1.75 t/m³
Standard Rock	1.95 t/m³ > loose density > 1.75 t/m³
Heavy Rock	Nominal loose density > 2.0 t/m³

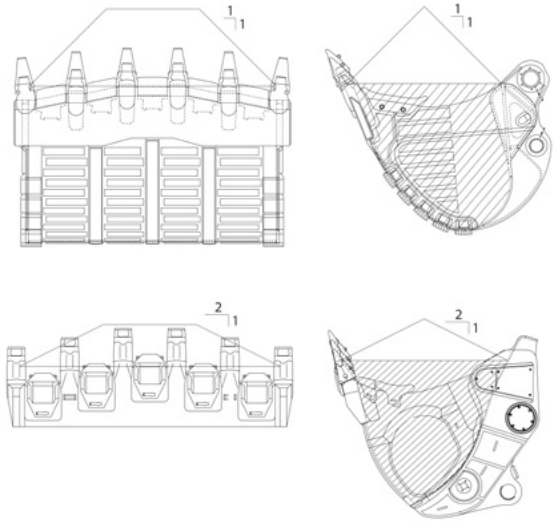
Coal buckets are suitable for material density up to 1.5 t/m³ (2,540 lb./yd³), depending on the configuration.

Hydraulic Mining Shovel backhoe and face shovel buckets use two additional descriptions of the wear package; Type and Duty.

Abrasion type wear packages use the hardest materials and are for excavated material that causes wear mainly through sliding.

Impact type wear packages use hard but tough materials and are for excavated material which is large and blocky in size.

The wear package also uses a “Duty” rating using three levels; “Extreme Duty”, “Heavy Duty” and “Standard Duty”. Extreme Duty wear packages provide the highest thickness and wear resistance but add the most weight to the bucket weight and reduce the available payload. Standard Duty wear packages provide the lowest resistance but add the least amount to the bucket weight.



Practical Bucket Capacity

The ratio of actual material volume or mass versus theoretical capacity is defined as the fill factor. Depending on the material, level of fragmentation, operating practice and machine setup, the actual fill factor with each pass can be more or less than 100%. An experienced, well-trained operator achieves between 90-95% fill factor in reasonably blasted material.

HMS PAYLOAD MANAGEMENT GUIDELINES

The total gross bucket weight and maximum material payload is defined as the Rated Swung Load (RSL). This value is published for each configuration of Hydraulic Mining Shovel. Refer to PEBJ0162 for further information.

Nominal Rated Payload is a reference value for a standard configuration with a loose material density of 1.8 t/m^3 (3,030 lb/yd^3).

Field Rated Payload is a specific calculation for a machine configuration including attachment, bucket and GET that can be used for productivity and truck match calculations for a unique customer application.

The Payload Management Guidelines allow buckets to be used with higher or lower density materials with varying fill factor results that still comply with the rated material payload and RSL. Rated payloads are published at 100% of allowed weight, even though a bucket can be filled beyond this and Caterpillar does allow up to 110% of Rated Payload on an infrequent basis.

Payloads of 80%-100% of Rated Payload result in higher productivity without reducing the life of structures and components. Excessive loads will likely reduce the life of structures and components thereby reducing the economic life of the machine.

Payloads from 100%-110% of Rated Payload can reduce life depending on the amount and duration of overloading.

Payloads between 100%-110% of Rated Payload should occur on no more than 10% of the loads.

The Maximum Allowable Payload is 110% of the Field Rated Payload.

CYCLE TIMES

The cycle time of a hydraulic mining shovel is an important productivity factor, and a key driver for high or low performance. There are two main areas that influence cycle times. One factor is the pure technical capability of the hydraulic mining shovel. The engine output and associated hydraulic power defines the maximum available oil flow, which consequently determines the speed of the hydraulic cylinders and swing drives, and ultimately the achievable working speed of the machine. The other factor is site specific parameters, such as operator skill, bench height (in particular for hydraulic front shovels), and swing angle, will impact cycle times considerably. Material fragmentation and penetration resistance make a big difference, as well.

Large hydraulic mining shovels are used as the prime mover in most applications. Therefore, the set-up of the loading area is pre-considered to be favorable for productive loading conditions when defining cycle times. Those are for example:

	For backhoe machines	For face shovel machines
Truck position	lower level	same level
Average swing angle	60°–90°	90°
Bench height	similar to stick length	75% of max. reach height

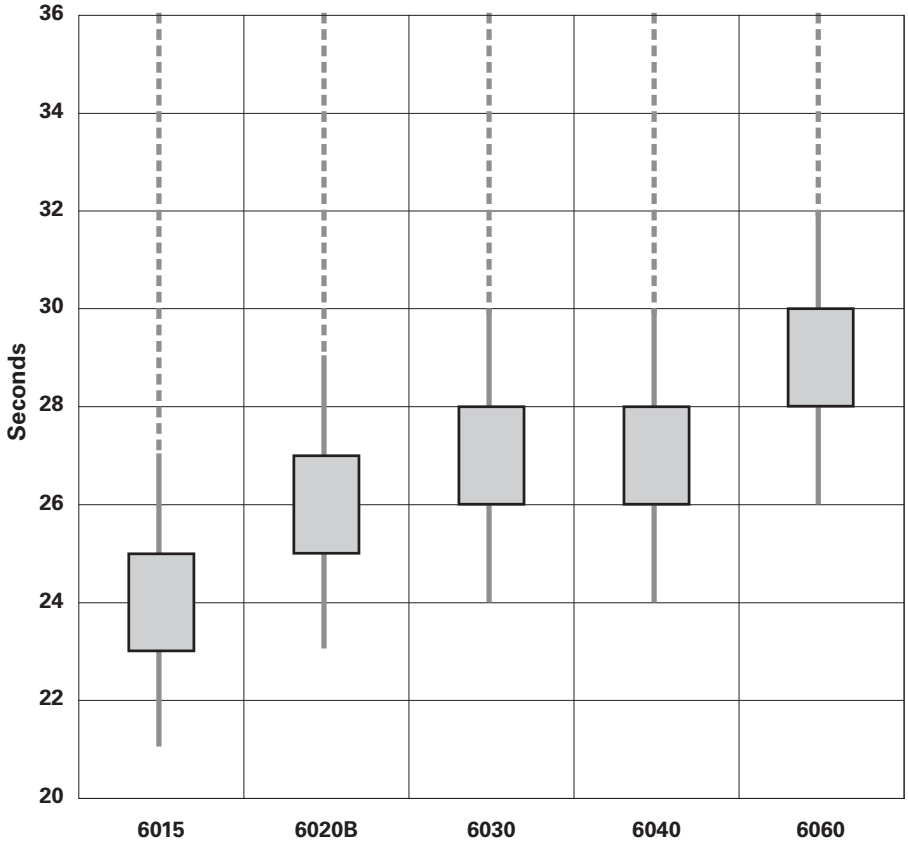
A complete working cycle of a hydraulic mining shovel can be separated into four segments:

- Bucket fill
- Swing loaded
- Dump
- Swing empty

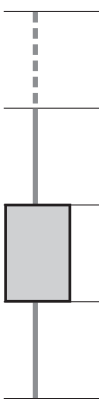
However, the bucket fill process constitutes the main fraction of the loading cycle, with a portion between 40% and 50% of the total time. This is heavily influenced by the material parameters, therefore, the different cycle times, shown for each model in the attached diagram, only consider the increasing effort to fill the bucket:

- **Ideal loading conditions** — Loose material, such as dry sand, soil and gravel, or re-handled or tipped material with no effort to achieve a good fill. Bucket can penetrate anywhere in the face or pile without resistance.
- **Average loading conditions** — Soft overburden with low grade of compaction, and well-shot material with good and consistent fragmentation. Operator doesn't have to concentrate on loosening or finding the right spot for penetration.
- **Fair digging conditions** — Shot rock with partially interlocked sections, or compacted overburden, requiring some effort to loosen the material.
- **Difficult digging conditions** — Inhomogeneous shot rock with some oversized boulders and interlocking, or unblasted material. Also free digging applications in sediment type of rock with fractures and layers which allow to penetrate the material. Considerable effort required to break out material.
- **Severe digging conditions** — Poorly shot rock with a large amount of oversized boulders and/or heavily interlocked material in the majority of the face. Also free digging operations with high digging resistance. In those conditions neither cycle times nor bucket fills are predictable.

Cycle Time Estimating Chart



Key



Severe digging conditions

Difficult digging conditions

Fair digging conditions

Average loading conditions

Ideal loading conditions

NOTE: The cycle times mentioned above are a guideline under the described application conditions.

Actual achievable cycle times depend on site specific conditions.

PRODUCTION OVERVIEWS

As with any other material loading machine, the productivity of a Hydraulic Mining Shovel depends on various aspects. These influencing factors are either defined by machine specifications, such as bucket capacity, or by application conditions and operator skills, such as bucket fill factor and cycle time. In addition, there are parameters, which fall neither under the hydrauilc mining shovel's physical capabilities, nor can they be controlled by the shovel operator or the set-up of the loading area. The major factors in this category are truck exchange time and overall mine efficiency.

The following tables indicate, for a typical hydraulic mining shovel/truck combination, the wide range of productivity figures under various conditions. The overall mine efficiency starts at 83%, which corresponds to 50 minutes per hour. This efficiency is generally achievable only during a short period of time, and could be the relevant factor for a short time production test. However, in the long-run, the overall mine efficiency tends to be in the 65% to 75% range. The grey highlighted area illustrates average productivity figures which can be expected under typical site conditions in well organized mines.

Furthermore, the size of the truck, and the required number of passes to load it, has an effect on hydraulic mining shovel productivity. The larger the truck, the higher the hydraulic mining shovel's productivity will be, because unproductive truck exchange time will be reduced.

The figures in the following tables given in t/h (tons/h) means the production per net loading hour and not per service meter units [SMU].

MODEL: 6015 ● Bucket Size: 8.1 m³ ● Number of Cycles: 4 ● Material Density: 1.80 t/m³

	Bucket Fill Factor		100%				95%			
	Load per Bucket		14.6 t				13.9 t			
	Load per Truck		58 t				55 t			
Overall Mine Efficiency	Cycle Time		21 sec	23 sec	25 sec	27 sec	21 sec	23 sec	25 sec	27 sec
	Truck Exchange		0.35 min	0.38 min	0.42 min	0.45 min	0.35 min	0.38 min	0.42 min	0.45 min
83% = maximum 50 min/hour	30 sec	0.50 min	1818 t/h	1721 t/h	1607 t/h	1531 t/h	1724 t/h	1633 t/h	1524 t/h	1452 t/h
	42 sec	0.70 min	1616 t/h	1539 t/h	1447 t/h	1385 t/h	1533 t/h	1460 t/h	1373 t/h	1314 t/h
	54 sec	0.90 min	1454 t/h	1392 t/h	1316 t/h	1264 t/h	1380 t/h	1320 t/h	1248 t/h	1200 t/h
	66 sec	1.10 min	1322 t/h	1270 t/h	1207 t/h	1163 t/h	1254 t/h	1205 t/h	1145 t/h	1104 t/h
80%	30 sec	0.50 min	1745 t/h	1652 t/h	1543 t/h	1470 t/h	1656 t/h	1568 t/h	1464 t/h	1394 t/h
	42 sec	0.70 min	1551 t/h	1478 t/h	1389 t/h	1330 t/h	1472 t/h	1402 t/h	1318 t/h	1262 t/h
	54 sec	0.90 min	1396 t/h	1336 t/h	1264 t/h	1214 t/h	1325 t/h	1268 t/h	1199 t/h	1152 t/h
	66 sec	1.10 min	1269 t/h	1220 t/h	1159 t/h	1117 t/h	1204 t/h	1157 t/h	1099 t/h	1060 t/h
75%	30 sec	0.50 min	1636 t/h	1549 t/h	1446 t/h	1378 t/h	1552 t/h	1469 t/h	1372 t/h	1307 t/h
	42 sec	0.70 min	1454 t/h	1385 t/h	1302 t/h	1246 t/h	1380 t/h	1314 t/h	1235 t/h	1182 t/h
	54 sec	0.90 min	1309 t/h	1252 t/h	1184 t/h	1138 t/h	1242 t/h	1188 t/h	1124 t/h	1080 t/h
	66 sec	1.10 min	1190 t/h	1143 t/h	1086 t/h	1047 t/h	1129 t/h	1084 t/h	1030 t/h	993 t/h
70%	30 sec	0.50 min	1526 t/h	1445 t/h	1349 t/h	1285 t/h	1448 t/h	1371 t/h	1280 t/h	1219 t/h
	42 sec	0.70 min	1357 t/h	1292 t/h	1215 t/h	1163 t/h	1287 t/h	1226 t/h	1153 t/h	1103 t/h
	54 sec	0.90 min	1221 t/h	1169 t/h	1105 t/h	1062 t/h	1158 t/h	1109 t/h	1048 t/h	1007 t/h
	66 sec	1.10 min	1110 t/h	1067 t/h	1013 t/h	977 t/h	1053 t/h	1012 t/h	961 t/h	927 t/h
65%	30 sec	0.50 min	1417 t/h	1341 t/h	1253 t/h	1193 t/h	1344 t/h	1273 t/h	1188 t/h	1132 t/h
	42 sec	0.70 min	1259 t/h	1200 t/h	1128 t/h	1080 t/h	1195 t/h	1138 t/h	1070 t/h	1024 t/h
	54 sec	0.90 min	1134 t/h	1085 t/h	1026 t/h	986 t/h	1075 t/h	1029 t/h	973 t/h	935 t/h
	66 sec	1.10 min	1030 t/h	990 t/h	941 t/h	907 t/h	978 t/h	939 t/h	892 t/h	860 t/h

	Bucket Fill Factor		90%				85%			
	Load per Bucket		13.1 t				12.4 t			
	Load per Truck		52 t				50 t			
Overall Mine Efficiency	Cycle Time		21 sec	23 sec	25 sec	27 sec	21 sec	23 sec	25 sec	27 sec
	Truck Exchange		0.35 min	0.38 min	0.42 min	0.45 min	0.35 min	0.38 min	0.42 min	0.45 min
83% = maximum 50 min/hour	30 sec	0.50 min	1634 t/h	1547 t/h	1444 t/h	1376 t/h	1543 t/h	1461 t/h	1364 t/h	1299 t/h
	42 sec	0.70 min	1452 t/h	1383 t/h	1300 t/h	1245 t/h	1371 t/h	1306 t/h	1228 t/h	1176 t/h
	54 sec	0.90 min	1307 t/h	1251 t/h	1183 t/h	1136 t/h	1234 t/h	1181 t/h	1117 t/h	1073 t/h
	66 sec	1.10 min	1188 t/h	1141 t/h	1085 t/h	1046 t/h	1122 t/h	1078 t/h	1024 t/h	987 t/h
80%	30 sec	0.50 min	1569 t/h	1485 t/h	1387 t/h	1321 t/h	1482 t/h	1403 t/h	1310 t/h	1248 t/h
	42 sec	0.70 min	1394 t/h	1328 t/h	1249 t/h	1195 t/h	1317 t/h	1254 t/h	1179 t/h	1129 t/h
	54 sec	0.90 min	1255 t/h	1201 t/h	1136 t/h	1091 t/h	1185 t/h	1134 t/h	1073 t/h	1031 t/h
	66 sec	1.10 min	1141 t/h	1096 t/h	1041 t/h	1004 t/h	1078 t/h	1035 t/h	984 t/h	948 t/h
75%	30 sec	0.50 min	1470 t/h	1392 t/h	1300 t/h	1238 t/h	1389 t/h	1315 t/h	1228 t/h	1169 t/h
	42 sec	0.70 min	1307 t/h	1245 t/h	1170 t/h	1120 t/h	1234 t/h	1176 t/h	1105 t/h	1058 t/h
	54 sec	0.90 min	1176 t/h	1126 t/h	1064 t/h	1023 t/h	1111 t/h	1063 t/h	1005 t/h	966 t/h
	66 sec	1.10 min	1069 t/h	1027 t/h	976 t/h	941 t/h	1010 t/h	970 t/h	922 t/h	889 t/h
70%	30 sec	0.50 min	1372 t/h	1299 t/h	1213 t/h	1155 t/h	1296 t/h	1227 t/h	1145 t/h	1091 t/h
	42 sec	0.70 min	1219 t/h	1161 t/h	1092 t/h	1045 t/h	1152 t/h	1097 t/h	1031 t/h	987 t/h
	54 sec	0.90 min	1098 t/h	1050 t/h	993 t/h	954 t/h	1037 t/h	992 t/h	938 t/h	901 t/h
	66 sec	1.10 min	998 t/h	959 t/h	911 t/h	878 t/h	942 t/h	905 t/h	860 t/h	829 t/h
65%	30 sec	0.50 min	1273 t/h	1206 t/h	1126 t/h	1072 t/h	1203 t/h	1139 t/h	1063 t/h	1013 t/h
	42 sec	0.70 min	1132 t/h	1078 t/h	1014 t/h	970 t/h	1069 t/h	1018 t/h	957 t/h	916 t/h
	54 sec	0.90 min	1019 t/h	975 t/h	922 t/h	886 t/h	962 t/h	921 t/h	871 t/h	837 t/h
	66 sec	1.10 min	926 t/h	890 t/h	845 t/h	815 t/h	875 t/h	840 t/h	799 t/h	770 t/h

MODEL: 6015 ● Bucket Size: 10.6 yd³ ● Number of Cycles: 4 ● Material Density: 3030 lb/yd³

	Bucket Fill Factor		100%				95%			
	Load per Bucket		16.1 tons				15.3 tons			
	Load per Truck		64 tons				61 tons			
Overall Mine Efficiency	Cycle Time		21 sec	23 sec	25 sec	27 sec	21 sec	23 sec	25 sec	27 sec
	Truck Exchange		0.35 min	0.38 min	0.42 min	0.45 min	0.35 min	0.38 min	0.42 min	0.45 min
83% = maximum 50 min/hour	30 sec	0.50 min	2004 tons/h	1897 tons/h	1771 tons/h	1687 tons/h	1901 tons/h	1800 tons/h	1680 tons/h	1601 tons/h
	42 sec	0.70 min	1781 tons/h	1696 tons/h	1595 tons/h	1527 tons/h	1690 tons/h	1609 tons/h	1513 tons/h	1448 tons/h
	54 sec	0.90 min	1603 tons/h	1534 tons/h	1451 tons/h	1394 tons/h	1521 tons/h	1455 tons/h	1376 tons/h	1322 tons/h
	66 sec	1.10 min	1457 tons/h	1400 tons/h	1330 tons/h	1282 tons/h	1382 tons/h	1328 tons/h	1262 tons/h	1217 tons/h
80%	30 sec	0.50 min	1924 tons/h	1822 tons/h	1701 tons/h	1620 tons/h	1825 tons/h	1728 tons/h	1614 tons/h	1537 tons/h
	42 sec	0.70 min	1710 tons/h	1629 tons/h	1532 tons/h	1466 tons/h	1622 tons/h	1545 tons/h	1453 tons/h	1391 tons/h
	54 sec	0.90 min	1539 tons/h	1473 tons/h	1393 tons/h	1338 tons/h	1460 tons/h	1397 tons/h	1321 tons/h	1270 tons/h
	66 sec	1.10 min	1399 tons/h	1344 tons/h	1277 tons/h	1231 tons/h	1327 tons/h	1275 tons/h	1212 tons/h	1168 tons/h
75%	30 sec	0.50 min	1803 tons/h	1707 tons/h	1594 tons/h	1519 tons/h	1711 tons/h	1620 tons/h	1512 tons/h	1441 tons/h
	42 sec	0.70 min	1603 tons/h	1527 tons/h	1435 tons/h	1374 tons/h	1521 tons/h	1448 tons/h	1362 tons/h	1303 tons/h
	54 sec	0.90 min	1443 tons/h	1381 tons/h	1306 tons/h	1254 tons/h	1369 tons/h	1310 tons/h	1239 tons/h	1190 tons/h
	66 sec	1.10 min	1311 tons/h	1260 tons/h	1197 tons/h	1154 tons/h	1244 tons/h	1195 tons/h	1136 tons/h	1095 tons/h
70%	30 sec	0.50 min	1683 tons/h	1593 tons/h	1487 tons/h	1417 tons/h	1596 tons/h	1511 tons/h	1411 tons/h	1344 tons/h
	42 sec	0.70 min	1496 tons/h	1424 tons/h	1339 tons/h	1282 tons/h	1419 tons/h	1351 tons/h	1271 tons/h	1216 tons/h
	54 sec	0.90 min	1346 tons/h	1288 tons/h	1218 tons/h	1171 tons/h	1277 tons/h	1222 tons/h	1156 tons/h	1110 tons/h
	66 sec	1.10 min	1224 tons/h	1176 tons/h	1117 tons/h	1077 tons/h	1161 tons/h	1115 tons/h	1060 tons/h	1022 tons/h
65%	30 sec	0.50 min	1562 tons/h	1479 tons/h	1381 tons/h	1315 tons/h	1482 tons/h	1403 tons/h	1310 tons/h	1248 tons/h
	42 sec	0.70 min	1388 tons/h	1322 tons/h	1243 tons/h	1190 tons/h	1317 tons/h	1254 tons/h	1180 tons/h	1129 tons/h
	54 sec	0.90 min	1250 tons/h	1196 tons/h	1131 tons/h	1087 tons/h	1185 tons/h	1134 tons/h	1073 tons/h	1031 tons/h
	66 sec	1.10 min	1136 tons/h	1091 tons/h	1037 tons/h	1000 tons/h	1078 tons/h	1035 tons/h	984 tons/h	948 tons/h

	Bucket Fill Factor		90%				85%			
	Load per Bucket		14.5 tons				13.7 tons			
	Load per Truck		58 tons				55 tons			
Overall Mine Efficiency	Cycle Time		21 sec	23 sec	25 sec	27 sec	21 sec	23 sec	25 sec	27 sec
	Truck Exchange		0.35 min	0.38 min	0.42 min	0.45 min	0.35 min	0.38 min	0.42 min	0.45 min
83% = maximum 50 min/hour	30 sec	0.50 min	1801 tons/h	1705 tons/h	1592 tons/h	1516 tons/h	1701 tons/h	1610 tons/h	1503 tons/h	1432 tons/h
	42 sec	0.70 min	1601 tons/h	1525 tons/h	1433 tons/h	1372 tons/h	1512 tons/h	1440 tons/h	1354 tons/h	1296 tons/h
	54 sec	0.90 min	1441 tons/h	1379 tons/h	1304 tons/h	1253 tons/h	1361 tons/h	1302 tons/h	1231 tons/h	1183 tons/h
	66 sec	1.10 min	1310 tons/h	1258 tons/h	1196 tons/h	1153 tons/h	1237 tons/h	1188 tons/h	1129 tons/h	1089 tons/h
80%	30 sec	0.50 min	1729 tons/h	1637 tons/h	1529 tons/h	1456 tons/h	1633 tons/h	1546 tons/h	1444 tons/h	1375 tons/h
	42 sec	0.70 min	1537 tons/h	1464 tons/h	1377 tons/h	1318 tons/h	1452 tons/h	1383 tons/h	1300 tons/h	1244 tons/h
	54 sec	0.90 min	1383 tons/h	1324 tons/h	1252 tons/h	1203 tons/h	1307 tons/h	1250 tons/h	1182 tons/h	1136 tons/h
	66 sec	1.10 min	1258 tons/h	1208 tons/h	1148 tons/h	1107 tons/h	1188 tons/h	1141 tons/h	1084 tons/h	1045 tons/h
75%	30 sec	0.50 min	1621 tons/h	1534 tons/h	1433 tons/h	1365 tons/h	1531 tons/h	1449 tons/h	1353 tons/h	1289 tons/h
	42 sec	0.70 min	1441 tons/h	1372 tons/h	1290 tons/h	1235 tons/h	1361 tons/h	1296 tons/h	1218 tons/h	1166 tons/h
	54 sec	0.90 min	1297 tons/h	1241 tons/h	1173 tons/h	1127 tons/h	1225 tons/h	1172 tons/h	1108 tons/h	1065 tons/h
	66 sec	1.10 min	1179 tons/h	1132 tons/h	1076 tons/h	1037 tons/h	1113 tons/h	1069 tons/h	1016 tons/h	980 tons/h
70%	30 sec	0.50 min	1512 tons/h	1432 tons/h	1337 tons/h	1273 tons/h	1428 tons/h	1352 tons/h	1263 tons/h	1203 tons/h
	42 sec	0.70 min	1344 tons/h	1280 tons/h	1204 tons/h	1152 tons/h	1270 tons/h	1209 tons/h	1137 tons/h	1088 tons/h
	54 sec	0.90 min	1210 tons/h	1158 tons/h	1095 tons/h	1052 tons/h	1143 tons/h	1093 tons/h	1034 tons/h	994 tons/h
	66 sec	1.10 min	1100 tons/h	1057 tons/h	1004 tons/h	968 tons/h	1039 tons/h	998 tons/h	948 tons/h	914 tons/h
65%	30 sec	0.50 min	1404 tons/h	1329 tons/h	1241 tons/h	1182 tons/h	1326 tons/h	1255 tons/h	1172 tons/h	1116 tons/h
	42 sec	0.70 min	1248 tons/h	1188 tons/h	1117 tons/h	1070 tons/h	1178 tons/h	1122 tons/h	1055 tons/h	1010 tons/h
	54 sec	0.90 min	1123 tons/h	1075 tons/h	1016 tons/h	977 tons/h	1061 tons/h	1015 tons/h	960 tons/h	922 tons/h
	66 sec	1.10 min	1021 tons/h	981 tons/h	932 tons/h	898 tons/h	964 tons/h	926 tons/h	880 tons/h	849 tons/h

MODEL: 6020B ● Bucket Size: 12.0 m³ ● Number of Cycles: 4 ● Material Density: 1.80 t/m³

	Bucket Fill Factor		100%				95%			
	Load per Bucket		21.6 t				20.5 t			
	Load per Truck		86 t				82 t			
Overall Mine Efficiency	Cycle Time		23 sec	25 sec	27 sec	29 sec	23 sec	25 sec	27 sec	29 sec
	Truck Exchange		0.38 min	0.42 min	0.45 min	0.48 min	0.38 min	0.42 min	0.45 min	0.48 min
83% = maximum 50 min/hour	30 sec	0.50 min	2546 t/h	2377 t/h	2265 t/h	2162 t/h	2419 t/h	2258 t/h	2151 t/h	2054 t/h
	42 sec	0.70 min	2277 t/h	2141 t/h	2049 t/h	1965 t/h	2163 t/h	2034 t/h	1946 t/h	1866 t/h
	54 sec	0.90 min	2059 t/h	1947 t/h	1871 t/h	1800 t/h	1956 t/h	1850 t/h	1777 t/h	1710 t/h
	66 sec	1.10 min	1879 t/h	1785 t/h	1721 t/h	1661 t/h	1785 t/h	1696 t/h	1635 t/h	1578 t/h
80%	30 sec	0.50 min	2445 t/h	2283 t/h	2175 t/h	2076 t/h	2323 t/h	2169 t/h	2066 t/h	1972 t/h
	42 sec	0.70 min	2186 t/h	2056 t/h	1967 t/h	1887 t/h	2077 t/h	1953 t/h	1869 t/h	1792 t/h
	54 sec	0.90 min	1977 t/h	1870 t/h	1796 t/h	1729 t/h	1878 t/h	1776 t/h	1707 t/h	1642 t/h
	66 sec	1.10 min	1804 t/h	1714 t/h	1653 t/h	1595 t/h	1714 t/h	1629 t/h	1570 t/h	1515 t/h
75%	30 sec	0.50 min	2291 t/h	2139 t/h	2038 t/h	1946 t/h	2177 t/h	2033 t/h	1936 t/h	1849 t/h
	42 sec	0.70 min	2049 t/h	1927 t/h	1844 t/h	1768 t/h	1946 t/h	1830 t/h	1752 t/h	1680 t/h
	54 sec	0.90 min	1853 t/h	1752 t/h	1684 t/h	1620 t/h	1760 t/h	1665 t/h	1599 t/h	1539 t/h
	66 sec	1.10 min	1691 t/h	1607 t/h	1549 t/h	1495 t/h	1606 t/h	1526 t/h	1472 t/h	1420 t/h
70%	30 sec	0.50 min	2138 t/h	1996 t/h	1902 t/h	1816 t/h	2031 t/h	1896 t/h	1807 t/h	1725 t/h
	42 sec	0.70 min	1912 t/h	1798 t/h	1721 t/h	1650 t/h	1816 t/h	1708 t/h	1635 t/h	1567 t/h
	54 sec	0.90 min	1729 t/h	1635 t/h	1571 t/h	1512 t/h	1642 t/h	1553 t/h	1492 t/h	1436 t/h
	66 sec	1.10 min	1578 t/h	1499 t/h	1445 t/h	1395 t/h	1499 t/h	1424 t/h	1373 t/h	1325 t/h
65%	30 sec	0.50 min	1985 t/h	1853 t/h	1765 t/h	1685 t/h	1885 t/h	1760 t/h	1677 t/h	1601 t/h
	42 sec	0.70 min	1775 t/h	1669 t/h	1597 t/h	1532 t/h	1686 t/h	1585 t/h	1517 t/h	1455 t/h
	54 sec	0.90 min	1605 t/h	1518 t/h	1458 t/h	1403 t/h	1525 t/h	1442 t/h	1385 t/h	1333 t/h
	66 sec	1.10 min	1465 t/h	1392 t/h	1342 t/h	1295 t/h	1391 t/h	1322 t/h	1275 t/h	1230 t/h

	Bucket Fill Factor		90%				85%			
	Load per Bucket		19.4 t				18.4 t			
	Load per Truck		78 t				73 t			
Overall Mine Efficiency	Cycle Time		23 sec	25 sec	27 sec	29 sec	23 sec	25 sec	27 sec	29 sec
	Truck Exchange		0.38 min	0.42 min	0.45 min	0.48 min	0.38 min	0.42 min	0.45 min	0.48 min
83% = maximum 50 min/hour	30 sec	0.50 min	2291 t/h	2139 t/h	2038 t/h	1946 t/h	2164 t/h	2021 t/h	1925 t/h	1838 t/h
	42 sec	0.70 min	2049 t/h	1927 t/h	1844 t/h	1768 t/h	1935 t/h	1820 t/h	1742 t/h	1670 t/h
	54 sec	0.90 min	1853 t/h	1752 t/h	1684 t/h	1620 t/h	1750 t/h	1655 t/h	1590 t/h	1530 t/h
	66 sec	1.10 min	1691 t/h	1607 t/h	1549 t/h	1495 t/h	1597 t/h	1518 t/h	1463 t/h	1412 t/h
80%	30 sec	0.50 min	2200 t/h	2054 t/h	1957 t/h	1869 t/h	2078 t/h	1940 t/h	1848 t/h	1765 t/h
	42 sec	0.70 min	1967 t/h	1850 t/h	1771 t/h	1698 t/h	1858 t/h	1747 t/h	1672 t/h	1604 t/h
	54 sec	0.90 min	1779 t/h	1683 t/h	1617 t/h	1556 t/h	1680 t/h	1589 t/h	1527 t/h	1469 t/h
	66 sec	1.10 min	1624 t/h	1543 t/h	1487 t/h	1436 t/h	1534 t/h	1457 t/h	1405 t/h	1356 t/h
75%	30 sec	0.50 min	2062 t/h	1926 t/h	1834 t/h	1751 t/h	1948 t/h	1819 t/h	1732 t/h	1654 t/h
	42 sec	0.70 min	1844 t/h	1734 t/h	1660 t/h	1591 t/h	1742 t/h	1638 t/h	1567 t/h	1503 t/h
	54 sec	0.90 min	1668 t/h	1577 t/h	1515 t/h	1458 t/h	1575 t/h	1489 t/h	1431 t/h	1377 t/h
	66 sec	1.10 min	1522 t/h	1446 t/h	1394 t/h	1346 t/h	1437 t/h	1366 t/h	1317 t/h	1271 t/h
70%	30 sec	0.50 min	1924 t/h	1797 t/h	1712 t/h	1634 t/h	1817 t/h	1697 t/h	1616 t/h	1543 t/h
	42 sec	0.70 min	1721 t/h	1618 t/h	1549 t/h	1485 t/h	1625 t/h	1528 t/h	1463 t/h	1402 t/h
	54 sec	0.90 min	1556 t/h	1471 t/h	1414 t/h	1361 t/h	1470 t/h	1390 t/h	1335 t/h	1285 t/h
	66 sec	1.10 min	1420 t/h	1349 t/h	1301 t/h	1256 t/h	1341 t/h	1274 t/h	1229 t/h	1186 t/h
65%	30 sec	0.50 min	1786 t/h	1668 t/h	1589 t/h	1517 t/h	1687 t/h	1575 t/h	1500 t/h	1433 t/h
	42 sec	0.70 min	1597 t/h	1502 t/h	1437 t/h	1378 t/h	1508 t/h	1418 t/h	1358 t/h	1302 t/h
	54 sec	0.90 min	1444 t/h	1366 t/h	1312 t/h	1263 t/h	1364 t/h	1290 t/h	1240 t/h	1193 t/h
	66 sec	1.10 min	1318 t/h	1253 t/h	1207 t/h	1165 t/h	1245 t/h	1183 t/h	1140 t/h	1101 t/h

MODEL: 6020B ● Bucket Size: 15.7 yd³ ● Number of Cycles: 4 ● Material Density: 3030 lb/yd³

Overall Mine Efficiency	Cycle Time	Truck Exchange	Bucket Fill Factor: 100%				Bucket Fill Factor: 95%			
			Load per Bucket: 23.8 tons				Load per Bucket: 22.6 tons			
			Load per Truck: 95 tons				Load per Truck: 90 tons			
	23 sec	**0.38 min**	2806 tons/h	2620 tons/h	2496 tons/h	2383 tons/h	2666 tons/h	2489 tons/h	2371 tons/h	2264 tons/h
83% = maximum 50 min/hour	42 sec	**0.70 min**	2509 tons/h	2360 tons/h	2259 tons/h	2166 tons/h	2384 tons/h	2242 tons/h	2146 tons/h	2057 tons/h
	54 sec	**0.90 min**	2269 tons/h	2146 tons/h	2062 tons/h	1984 tons/h	2156 tons/h	2039 tons/h	1959 tons/h	1885 tons/h
	66 sec	**1.10 min**	2071 tons/h	1968 tons/h	1897 tons/h	1831 tons/h	1968 tons/h	1870 tons/h	1802 tons/h	1740 tons/h
	30 sec	**0.50 min**	2695 tons/h	2516 tons/h	2397 tons/h	2289 tons/h	2560 tons/h	2390 tons/h	2277 tons/h	2174 tons/h
80%	42 sec	**0.70 min**	2410 tons/h	2266 tons/h	2169 tons/h	2080 tons/h	2289 tons/h	2153 tons/h	2060 tons/h	1976 tons/h
	54 sec	**0.90 min**	2179 tons/h	2061 tons/h	1980 tons/h	1906 tons/h	2070 tons/h	1958 tons/h	1881 tons/h	1810 tons/h
	66 sec	**1.10 min**	1989 tons/h	1890 tons/h	1822 tons/h	1758 tons/h	1889 tons/h	1795 tons/h	1731 tons/h	1671 tons/h
	30 sec	**0.50 min**	2526 tons/h	2358 tons/h	2247 tons/h	2145 tons/h	2400 tons/h	2240 tons/h	2134 tons/h	2038 tons/h
75%	42 sec	**0.70 min**	2259 tons/h	2124 tons/h	2033 tons/h	1949 tons/h	2146 tons/h	2018 tons/h	1931 tons/h	1852 tons/h
	54 sec	**0.90 min**	2042 tons/h	1932 tons/h	1856 tons/h	1786 tons/h	1940 tons/h	1835 tons/h	1763 tons/h	1697 tons/h
	66 sec	**1.10 min**	1864 tons/h	1771 tons/h	1707 tons/h	1648 tons/h	1771 tons/h	1683 tons/h	1622 tons/h	1566 tons/h
	30 sec	**0.50 min**	2357 tons/h	2201 tons/h	2096 tons/h	2001 tons/h	2239 tons/h	2090 tons/h	1991 tons/h	1901 tons/h
70%	42 sec	**0.70 min**	2107 tons/h	1982 tons/h	1897 tons/h	1819 tons/h	2002 tons/h	1882 tons/h	1802 tons/h	1728 tons/h
	54 sec	**0.90 min**	1906 tons/h	1802 tons/h	1732 tons/h	1666 tons/h	1810 tons/h	1712 tons/h	1645 tons/h	1583 tons/h
	66 sec	**1.10 min**	1739 tons/h	1653 tons/h	1593 tons/h	1538 tons/h	1652 tons/h	1570 tons/h	1514 tons/h	1461 tons/h
	30 sec	**0.50 min**	2188 tons/h	2043 tons/h	1946 tons/h	1858 tons/h	2078 tons/h	1941 tons/h	1849 tons/h	1765 tons/h
65%	42 sec	**0.70 min**	1956 tons/h	1839 tons/h	1761 tons/h	1688 tons/h	1858 tons/h	1747 tons/h	1673 tons/h	1604 tons/h
	54 sec	**0.90 min**	1769 tons/h	1673 tons/h	1607 tons/h	1547 tons/h	1681 tons/h	1589 tons/h	1527 tons/h	1470 tons/h
	66 sec	**1.10 min**	1614 tons/h	1534 tons/h	1479 tons/h	1427 tons/h	1534 tons/h	1457 tons/h	1405 tons/h	1356 tons/h

Overall Mine Efficiency	Cycle Time	Truck Exchange	Bucket Fill Factor: 90%				Bucket Fill Factor: 85%			
			Load per Bucket: 21.4 tons				Load per Bucket: 20.2 tons			
			Load per Truck: 86 tons				Load per Truck: 81 tons			
	23 sec	**0.38 min**	2526 tons/h	2358 tons/h	2247 tons/h	2145 tons/h	2386 tons/h	2227 tons/h	2122 tons/h	2026 tons/h
83% = maximum 50 min/hour	42 sec	**0.70 min**	2259 tons/h	2124 tons/h	2033 tons/h	1949 tons/h	2133 tons/h	2006 tons/h	1920 tons/h	1841 tons/h
	54 sec	**0.90 min**	2042 tons/h	1932 tons/h	1856 tons/h	1786 tons/h	1929 tons/h	1824 tons/h	1753 tons/h	1687 tons/h
	66 sec	**1.10 min**	1864 tons/h	1771 tons/h	1707 tons/h	1648 tons/h	1760 tons/h	1673 tons/h	1613 tons/h	1557 tons/h
	30 sec	**0.50 min**	2425 tons/h	2265 tons/h	2157 tons/h	2060 tons/h	2291 tons/h	2139 tons/h	2037 tons/h	1945 tons/h
80%	42 sec	**0.70 min**	2169 tons/h	2039 tons/h	1952 tons/h	1872 tons/h	2048 tons/h	1926 tons/h	1843 tons/h	1768 tons/h
	54 sec	**0.90 min**	1961 tons/h	1855 tons/h	1782 tons/h	1715 tons/h	1852 tons/h	1752 tons/h	1683 tons/h	1620 tons/h
	66 sec	**1.10 min**	1790 tons/h	1701 tons/h	1640 tons/h	1583 tons/h	1690 tons/h	1606 tons/h	1548 tons/h	1495 tons/h
	30 sec	**0.50 min**	2273 tons/h	2123 tons/h	2022 tons/h	1931 tons/h	2147 tons/h	2005 tons/h	1910 tons/h	1823 tons/h
75%	42 sec	**0.70 min**	2033 tons/h	1911 tons/h	1829 tons/h	1754 tons/h	1920 tons/h	1805 tons/h	1728 tons/h	1657 tons/h
	54 sec	**0.90 min**	1838 tons/h	1738 tons/h	1670 tons/h	1607 tons/h	1736 tons/h	1642 tons/h	1578 tons/h	1518 tons/h
	66 sec	**1.10 min**	1678 tons/h	1594 tons/h	1537 tons/h	1483 tons/h	1584 tons/h	1506 tons/h	1451 tons/h	1401 tons/h
	30 sec	**0.50 min**	2121 tons/h	1980 tons/h	1887 tons/h	1801 tons/h	2003 tons/h	1870 tons/h	1782 tons/h	1701 tons/h
70%	42 sec	**0.70 min**	1897 tons/h	1783 tons/h	1707 tons/h	1637 tons/h	1791 tons/h	1684 tons/h	1612 tons/h	1546 tons/h
	54 sec	**0.90 min**	1715 tons/h	1622 tons/h	1559 tons/h	1500 tons/h	1620 tons/h	1532 tons/h	1472 tons/h	1417 tons/h
	66 sec	**1.10 min**	1565 tons/h	1487 tons/h	1434 tons/h	1384 tons/h	1478 tons/h	1405 tons/h	1354 tons/h	1307 tons/h
	30 sec	**0.50 min**	1969 tons/h	1838 tons/h	1751 tons/h	1672 tons/h	1860 tons/h	1736 tons/h	1654 tons/h	1579 tons/h
65%	42 sec	**0.70 min**	1761 tons/h	1655 tons/h	1585 tons/h	1519 tons/h	1663 tons/h	1563 tons/h	1496 tons/h	1435 tons/h
	54 sec	**0.90 min**	1592 tons/h	1506 tons/h	1447 tons/h	1392 tons/h	1504 tons/h	1422 tons/h	1366 tons/h	1315 tons/h
	66 sec	**1.10 min**	1453 tons/h	1381 tons/h	1331 tons/h	1285 tons/h	1372 tons/h	1304 tons/h	1257 tons/h	1213 tons/h

MODEL: 6030 with Backhoe attachment ● Bucket Size: 17.0 m³ ● Number of Cycles: 5 ● Material Density: 1.80 t/m³

	Bucket Fill Factor		100%				95%			
	Load per Bucket		30.6 t				29.1 t			
	Load per Truck		153 t				145 t			
Overall Mine Efficiency	Cycle Time	24 sec	26 sec	28 sec	30 sec	24 sec	26 sec	28 sec	30 sec
	Truck Exchange	0.40 min	0.43 min	0.47 min	0.50 min	0.40 min	0.43 min	0.47 min	0.50 min
83% = maximum 50 min/hour	30 sec	0.50 min	3544 t/h	3357 t/h	3136 t/h	2988 t/h	3367 t/h	3189 t/h	2979 t/h	2839 t/h
	42 sec	0.70 min	3242 t/h	3085 t/h	2897 t/h	2771 t/h	3080 t/h	2931 t/h	2752 t/h	2632 t/h
	54 sec	0.90 min	2988 t/h	2854 t/h	2692 t/h	2583 t/h	2839 t/h	2711 t/h	2558 t/h	2454 t/h
	66 sec	1.10 min	2771 t/h	2655 t/h	2515 t/h	2419 t/h	2632 t/h	2522 t/h	2389 t/h	2298 t/h
80%	30 sec	0.50 min	3403 t/h	3223 t/h	3011 t/h	2869 t/h	3233 t/h	3062 t/h	2860 t/h	2726 t/h
	42 sec	0.70 min	3113 t/h	2962 t/h	2782 t/h	2661 t/h	2958 t/h	2814 t/h	2643 t/h	2528 t/h
	54 sec	0.90 min	2869 t/h	2740 t/h	2585 t/h	2480 t/h	2726 t/h	2603 t/h	2456 t/h	2356 t/h
	66 sec	1.10 min	2661 t/h	2549 t/h	2415 t/h	2323 t/h	2528 t/h	2422 t/h	2294 t/h	2207 t/h
75%	30 sec	0.50 min	3190 t/h	3021 t/h	2822 t/h	2689 t/h	3030 t/h	2870 t/h	2681 t/h	2555 t/h
	42 sec	0.70 min	2918 t/h	2776 t/h	2607 t/h	2494 t/h	2772 t/h	2637 t/h	2477 t/h	2369 t/h
	54 sec	0.90 min	2689 t/h	2568 t/h	2423 t/h	2325 t/h	2555 t/h	2440 t/h	2302 t/h	2208 t/h
	66 sec	1.10 min	2494 t/h	2389 t/h	2263 t/h	2177 t/h	2369 t/h	2270 t/h	2150 t/h	2068 t/h
70%	30 sec	0.50 min	2976 t/h	2819 t/h	2633 t/h	2509 t/h	2827 t/h	2678 t/h	2501 t/h	2384 t/h
	42 sec	0.70 min	2723 t/h	2590 t/h	2433 t/h	2327 t/h	2587 t/h	2461 t/h	2311 t/h	2210 t/h
	54 sec	0.90 min	2509 t/h	2396 t/h	2261 t/h	2169 t/h	2384 t/h	2277 t/h	2148 t/h	2061 t/h
	66 sec	1.10 min	2327 t/h	2229 t/h	2112 t/h	2031 t/h	2210 t/h	2118 t/h	2006 t/h	1930 t/h
65%	30 sec	0.50 min	2763 t/h	2617 t/h	2444 t/h	2329 t/h	2624 t/h	2486 t/h	2322 t/h	2213 t/h
	42 sec	0.70 min	2527 t/h	2405 t/h	2258 t/h	2160 t/h	2401 t/h	2284 t/h	2145 t/h	2052 t/h
	54 sec	0.90 min	2329 t/h	2225 t/h	2099 t/h	2013 t/h	2213 t/h	2113 t/h	1994 t/h	1913 t/h
	66 sec	1.10 min	2160 t/h	2069 t/h	1960 t/h	1886 t/h	2052 t/h	1966 t/h	1862 t/h	1791 t/h

	Bucket Fill Factor		90%				85%			
	Load per Bucket		27.5 t				26.0 t			
	Load per Truck		138 t				130 t			
Overall Mine Efficiency	Cycle Time	24 sec	26 sec	28 sec	30 sec	24 sec	26 sec	28 sec	30 sec
	Truck Exchange	0.40 min	0.43 min	0.47 min	0.50 min	0.40 min	0.43 min	0.47 min	0.50 min
83% = maximum 50 min/hour	30 sec	0.50 min	3190 t/h	3021 t/h	2822 t/h	2689 t/h	3012 t/h	2853 t/h	2665 t/h	2540 t/h
	42 sec	0.70 min	2918 t/h	2776 t/h	2607 t/h	2494 t/h	2756 t/h	2622 t/h	2463 t/h	2355 t/h
	54 sec	0.90 min	2689 t/h	2568 t/h	2423 t/h	2325 t/h	2540 t/h	2426 t/h	2289 t/h	2195 t/h
	66 sec	1.10 min	2494 t/h	2389 t/h	2263 t/h	2177 t/h	2355 t/h	2257 t/h	2137 t/h	2056 t/h
80%	30 sec	0.50 min	3063 t/h	2901 t/h	2710 t/h	2582 t/h	2893 t/h	2740 t/h	2559 t/h	2439 t/h
	42 sec	0.70 min	2802 t/h	2666 t/h	2504 t/h	2394 t/h	2646 t/h	2518 t/h	2365 t/h	2261 t/h
	54 sec	0.90 min	2582 t/h	2466 t/h	2327 t/h	2232 t/h	2439 t/h	2329 t/h	2198 t/h	2108 t/h
	66 sec	1.10 min	2394 t/h	2294 t/h	2173 t/h	2090 t/h	2261 t/h	2167 t/h	2052 t/h	1974 t/h
75%	30 sec	0.50 min	2871 t/h	2719 t/h	2540 t/h	2420 t/h	2711 t/h	2568 t/h	2399 t/h	2286 t/h
	42 sec	0.70 min	2626 t/h	2499 t/h	2347 t/h	2244 t/h	2480 t/h	2360 t/h	2216 t/h	2120 t/h
	54 sec	0.90 min	2420 t/h	2312 t/h	2181 t/h	2092 t/h	2286 t/h	2183 t/h	2060 t/h	1976 t/h
	66 sec	1.10 min	2244 t/h	2150 t/h	2037 t/h	1959 t/h	2120 t/h	2031 t/h	1924 t/h	1850 t/h
70%	30 sec	0.50 min	2678 t/h	2537 t/h	2370 t/h	2258 t/h	2530 t/h	2396 t/h	2238 t/h	2133 t/h
	42 sec	0.70 min	2450 t/h	2331 t/h	2190 t/h	2094 t/h	2314 t/h	2202 t/h	2068 t/h	1978 t/h
	54 sec	0.90 min	2258 t/h	2157 t/h	2035 t/h	1952 t/h	2133 t/h	2037 t/h	1922 t/h	1844 t/h
	66 sec	1.10 min	2094 t/h	2006 t/h	1901 t/h	1828 t/h	1978 t/h	1895 t/h	1795 t/h	1727 t/h
65%	30 sec	0.50 min	2486 t/h	2355 t/h	2200 t/h	2096 t/h	2348 t/h	2224 t/h	2078 t/h	1980 t/h
	42 sec	0.70 min	2275 t/h	2164 t/h	2033 t/h	1944 t/h	2148 t/h	2044 t/h	1920 t/h	1836 t/h
	54 sec	0.90 min	2096 t/h	2002 t/h	1889 t/h	1812 t/h	1980 t/h	1891 t/h	1784 t/h	1711 t/h
	66 sec	1.10 min	1944 t/h	1863 t/h	1764 t/h	1697 t/h	1836 t/h	1759 t/h	1666 t/h	1603 t/h

MODEL: 6030 with Backhoe attachment ● **Bucket Size: 22.2 yd³** ● **Number of Cycles: 5** ● **Material Density: 3030 lb/yd³**

	Bucket Fill Factor		100%				95%			
	Load per Bucket		33.7 tons				32.1 tons			
	Load per Truck		169 tons				160 tons			
Overall Mine Efficiency	Cycle Time		24 sec	26 sec	28 sec	30 sec	24 sec	26 sec	28 sec	30 sec
	Truck Exchange		0.40 min	0.43 min	0.47 min	0.50 min	0.40 min	0.43 min	0.47 min	0.50 min
83% = maximum 50 min/hour	30 sec	0.50 min	3906 tons/h	3700 tons/h	3456 tons/h	3294 tons/h	3715 tons/h	3519 tons/h	3287 tons/h	3132 tons/h
	42 sec	0.70 min	3574 tons/h	3400 tons/h	3194 tons/h	3054 tons/h	3399 tons/h	3234 tons/h	3037 tons/h	2904 tons/h
	54 sec	0.90 min	3294 tons/h	3146 tons/h	2968 tons/h	2847 tons/h	3132 tons/h	2991 tons/h	2822 tons/h	2708 tons/h
	66 sec	1.10 min	3054 tons/h	2926 tons/h	2772 tons/h	2666 tons/h	2904 tons/h	2783 tons/h	2636 tons/h	2536 tons/h
80%	30 sec	0.50 min	3751 tons/h	3553 tons/h	3319 tons/h	3163 tons/h	3567 tons/h	3379 tons/h	3156 tons/h	3008 tons/h
	42 sec	0.70 min	3432 tons/h	3265 tons/h	3067 tons/h	2933 tons/h	3264 tons/h	3105 tons/h	2916 tons/h	2789 tons/h
	54 sec	0.90 min	3163 tons/h	3021 tons/h	2850 tons/h	2734 tons/h	3008 tons/h	2873 tons/h	2710 tons/h	2600 tons/h
	66 sec	1.10 min	2933 tons/h	2810 tons/h	2662 tons/h	2560 tons/h	2789 tons/h	2672 tons/h	2531 tons/h	2435 tons/h
75%	30 sec	0.50 min	3516 tons/h	3330 tons/h	3111 tons/h	2964 tons/h	3343 tons/h	3167 tons/h	2958 tons/h	2819 tons/h
	42 sec	0.70 min	3217 tons/h	3060 tons/h	2874 tons/h	2749 tons/h	3059 tons/h	2910 tons/h	2733 tons/h	2614 tons/h
	54 sec	0.90 min	2964 tons/h	2831 tons/h	2671 tons/h	2562 tons/h	2819 tons/h	2692 tons/h	2540 tons/h	2437 tons/h
	66 sec	1.10 min	2749 tons/h	2634 tons/h	2495 tons/h	2400 tons/h	2614 tons/h	2505 tons/h	2372 tons/h	2282 tons/h
70%	30 sec	0.50 min	3281 tons/h	3107 tons/h	2903 tons/h	2766 tons/h	3120 tons/h	2955 tons/h	2760 tons/h	2630 tons/h
	42 sec	0.70 min	3001 tons/h	2856 tons/h	2682 tons/h	2565 tons/h	2854 tons/h	2716 tons/h	2550 tons/h	2439 tons/h
	54 sec	0.90 min	2766 tons/h	2642 tons/h	2492 tons/h	2391 tons/h	2630 tons/h	2512 tons/h	2370 tons/h	2274 tons/h
	66 sec	1.10 min	2565 tons/h	2458 tons/h	2328 tons/h	2239 tons/h	2439 tons/h	2337 tons/h	2214 tons/h	2129 tons/h
65%	30 sec	0.50 min	3045 tons/h	2884 tons/h	2694 tons/h	2568 tons/h	2896 tons/h	2743 tons/h	2562 tons/h	2442 tons/h
	42 sec	0.70 min	2786 tons/h	2651 tons/h	2489 tons/h	2381 tons/h	2649 tons/h	2521 tons/h	2367 tons/h	2264 tons/h
	54 sec	0.90 min	2568 tons/h	2452 tons/h	2313 tons/h	2219 tons/h	2442 tons/h	2332 tons/h	2200 tons/h	2111 tons/h
	66 sec	1.10 min	2381 tons/h	2281 tons/h	2161 tons/h	2078 tons/h	2264 tons/h	2169 tons/h	2055 tons/h	1977 tons/h

	Bucket Fill Factor		90%				85%			
	Load per Bucket		30.3 tons				28.7 tons			
	Load per Truck		152 tons				143 tons			
Overall Mine Efficiency	Cycle Time		24 sec	26 sec	28 sec	30 sec	24 sec	26 sec	28 sec	30 sec
	Truck Exchange		0.40 min	0.43 min	0.47 min	0.50 min	0.40 min	0.43 min	0.47 min	0.50 min
83% = maximum 50 min/hour	30 sec	0.50 min	3511 tons/h	3325 tons/h	3106 tons/h	2960 tons/h	3319 tons/h	3144 tons/h	2937 tons/h	2799 tons/h
	42 sec	0.70 min	3212 tons/h	3056 tons/h	2870 tons/h	2745 tons/h	3037 tons/h	2889 tons/h	2713 tons/h	2595 tons/h
	54 sec	0.90 min	2960 tons/h	2827 tons/h	2667 tons/h	2559 tons/h	2799 tons/h	2673 tons/h	2522 tons/h	2419 tons/h
	66 sec	1.10 min	2745 tons/h	2630 tons/h	2491 tons/h	2396 tons/h	2595 tons/h	2487 tons/h	2355 tons/h	2266 tons/h
80%	30 sec	0.50 min	3371 tons/h	3193 tons/h	2983 tons/h	2842 tons/h	3187 tons/h	3019 tons/h	2820 tons/h	2687 tons/h
	42 sec	0.70 min	3084 tons/h	2934 tons/h	2756 tons/h	2636 tons/h	2916 tons/h	2774 tons/h	2606 tons/h	2492 tons/h
	54 sec	0.90 min	2842 tons/h	2715 tons/h	2561 tons/h	2457 tons/h	2687 tons/h	2567 tons/h	2421 tons/h	2323 tons/h
	66 sec	1.10 min	2636 tons/h	2525 tons/h	2392 tons/h	2301 tons/h	2492 tons/h	2388 tons/h	2262 tons/h	2175 tons/h
75%	30 sec	0.50 min	3160 tons/h	2993 tons/h	2796 tons/h	2664 tons/h	2987 tons/h	2829 tons/h	2643 tons/h	2519 tons/h
	42 sec	0.70 min	2891 tons/h	2750 tons/h	2583 tons/h	2470 tons/h	2733 tons/h	2600 tons/h	2442 tons/h	2336 tons/h
	54 sec	0.90 min	2664 tons/h	2544 tons/h	2400 tons/h	2303 tons/h	2519 tons/h	2406 tons/h	2270 tons/h	2177 tons/h
	66 sec	1.10 min	2470 tons/h	2367 tons/h	2242 tons/h	2157 tons/h	2336 tons/h	2238 tons/h	2120 tons/h	2039 tons/h
70%	30 sec	0.50 min	2948 tons/h	2792 tons/h	2608 tons/h	2486 tons/h	2787 tons/h	2640 tons/h	2466 tons/h	2350 tons/h
	42 sec	0.70 min	2697 tons/h	2566 tons/h	2410 tons/h	2305 tons/h	2550 tons/h	2426 tons/h	2279 tons/h	2179 tons/h
	54 sec	0.90 min	2486 tons/h	2374 tons/h	2240 tons/h	2149 tons/h	2350 tons/h	2245 tons/h	2118 tons/h	2031 tons/h
	66 sec	1.10 min	2305 tons/h	2209 tons/h	2092 tons/h	2012 tons/h	2179 tons/h	2088 tons/h	1978 tons/h	1902 tons/h
65%	30 sec	0.50 min	2737 tons/h	2592 tons/h	2421 tons/h	2307 tons/h	2587 tons/h	2451 tons/h	2289 tons/h	2182 tons/h
	42 sec	0.70 min	2504 tons/h	2382 tons/h	2237 tons/h	2140 tons/h	2367 tons/h	2252 tons/h	2115 tons/h	2023 tons/h
	54 sec	0.90 min	2307 tons/h	2204 tons/h	2079 tons/h	1995 tons/h	2182 tons/h	2083 tons/h	1966 tons/h	1886 tons/h
	66 sec	1.10 min	2140 tons/h	2050 tons/h	1942 tons/h	1868 tons/h	2023 tons/h	1938 tons/h	1836 tons/h	1766 tons/h

MODEL: 6030 with Face Shovel attachment ● Bucket Size: 16.5 m³ ● Number of Cycles: 5 ● Material Density: 1.80 t/m³

	Bucket Fill Factor		100%				95%			
	Load per Bucket		29.7 t				28.2 t			
	Load per Truck		149 t				141 t			
Overall Mine Efficiency	Cycle Time		24 sec	26 sec	28 sec	30 sec	24 sec	26 sec	28 sec	30 sec
	Truck Exchange		0.40 min	0.43 min	0.47 min	0.50 min	0.40 min	0.43 min	0.47 min	0.50 min
83% = maximum 50 min/hour	30 sec	0.50 min	3440 t/h	3258 t/h	3043 t/h	2900 t/h	3266 t/h	3093 t/h	2890 t/h	2754 t/h
	42 sec	0.70 min	3147 t/h	2994 t/h	2812 t/h	2689 t/h	2988 t/h	2843 t/h	2670 t/h	2553 t/h
	54 sec	0.90 min	2900 t/h	2770 t/h	2613 t/h	2507 t/h	2754 t/h	2630 t/h	2481 t/h	2380 t/h
	66 sec	1.10 min	2689 t/h	2577 t/h	2441 t/h	2348 t/h	2553 t/h	2447 t/h	2317 t/h	2229 t/h
80%	30 sec	0.50 min	3303 t/h	3128 t/h	2922 t/h	2785 t/h	3136 t/h	2970 t/h	2775 t/h	2644 t/h
	42 sec	0.70 min	3022 t/h	2875 t/h	2700 t/h	2582 t/h	2869 t/h	2730 t/h	2564 t/h	2452 t/h
	54 sec	0.90 min	2785 t/h	2660 t/h	2509 t/h	2407 t/h	2644 t/h	2525 t/h	2383 t/h	2286 t/h
	66 sec	1.10 min	2582 t/h	2474 t/h	2344 t/h	2254 t/h	2452 t/h	2349 t/h	2225 t/h	2141 t/h
75%	30 sec	0.50 min	3096 t/h	2932 t/h	2739 t/h	2610 t/h	2939 t/h	2784 t/h	2601 t/h	2478 t/h
	42 sec	0.70 min	2832 t/h	2695 t/h	2531 t/h	2420 t/h	2689 t/h	2559 t/h	2403 t/h	2298 t/h
	54 sec	0.90 min	2610 t/h	2493 t/h	2352 t/h	2256 t/h	2478 t/h	2367 t/h	2233 t/h	2142 t/h
	66 sec	1.10 min	2420 t/h	2319 t/h	2197 t/h	2113 t/h	2298 t/h	2202 t/h	2086 t/h	2006 t/h
70%	30 sec	0.50 min	2888 t/h	2736 t/h	2556 t/h	2435 t/h	2743 t/h	2598 t/h	2427 t/h	2312 t/h
	42 sec	0.70 min	2643 t/h	2514 t/h	2361 t/h	2258 t/h	2509 t/h	2387 t/h	2242 t/h	2144 t/h
	54 sec	0.90 min	2435 t/h	2326 t/h	2194 t/h	2105 t/h	2312 t/h	2208 t/h	2084 t/h	1999 t/h
	66 sec	1.10 min	2258 t/h	2164 t/h	2050 t/h	1972 t/h	2144 t/h	2055 t/h	1946 t/h	1872 t/h
65%	30 sec	0.50 min	2681 t/h	2540 t/h	2372 t/h	2261 t/h	2546 t/h	2411 t/h	2253 t/h	2147 t/h
	42 sec	0.70 min	2453 t/h	2334 t/h	2192 t/h	2096 t/h	2329 t/h	2216 t/h	2081 t/h	1990 t/h
	54 sec	0.90 min	2261 t/h	2159 t/h	2037 t/h	1954 t/h	2147 t/h	2050 t/h	1934 t/h	1855 t/h
	66 sec	1.10 min	2096 t/h	2009 t/h	1903 t/h	1830 t/h	1990 t/h	1907 t/h	1806 t/h	1738 t/h

	Bucket Fill Factor		90%				85%			
	Load per Bucket		26.7 t				25.2 t			
	Load per Truck		134 t				126 t			
Overall Mine Efficiency	Cycle Time		24 sec	26 sec	28 sec	30 sec	24 sec	26 sec	28 sec	30 sec
	Truck Exchange		0.40 min	0.43 min	0.47 min	0.50 min	0.40 min	0.43 min	0.47 min	0.50 min
83% = maximum 50 min/hour	30 sec	0.50 min	3092 t/h	2929 t/h	2736 t/h	2607 t/h	2919 t/h	2764 t/h	2582 t/h	2461 t/h
	42 sec	0.70 min	2829 t/h	2692 t/h	2528 t/h	2418 t/h	2670 t/h	2540 t/h	2386 t/h	2282 t/h
	54 sec	0.90 min	2607 t/h	2490 t/h	2349 t/h	2254 t/h	2461 t/h	2350 t/h	2217 t/h	2127 t/h
	66 sec	1.10 min	2418 t/h	2316 t/h	2194 t/h	2111 t/h	2282 t/h	2186 t/h	2071 t/h	1992 t/h
80%	30 sec	0.50 min	2969 t/h	2812 t/h	2627 t/h	2504 t/h	2802 t/h	2654 t/h	2480 t/h	2363 t/h
	42 sec	0.70 min	2717 t/h	2585 t/h	2427 t/h	2321 t/h	2564 t/h	2439 t/h	2291 t/h	2191 t/h
	54 sec	0.90 min	2504 t/h	2391 t/h	2256 t/h	2164 t/h	2363 t/h	2257 t/h	2129 t/h	2042 t/h
	66 sec	1.10 min	2321 t/h	2224 t/h	2107 t/h	2027 t/h	2191 t/h	2099 t/h	1989 t/h	1913 t/h
75%	30 sec	0.50 min	2783 t/h	2636 t/h	2462 t/h	2346 t/h	2627 t/h	2488 t/h	2324 t/h	2215 t/h
	42 sec	0.70 min	2546 t/h	2422 t/h	2275 t/h	2176 t/h	2403 t/h	2286 t/h	2147 t/h	2054 t/h
	54 sec	0.90 min	2346 t/h	2241 t/h	2114 t/h	2028 t/h	2215 t/h	2115 t/h	1996 t/h	1914 t/h
	66 sec	1.10 min	2176 t/h	2085 t/h	1975 t/h	1900 t/h	2054 t/h	1968 t/h	1864 t/h	1793 t/h
70%	30 sec	0.50 min	2597 t/h	2459 t/h	2298 t/h	2189 t/h	2451 t/h	2321 t/h	2168 t/h	2066 t/h
	42 sec	0.70 min	2376 t/h	2260 t/h	2123 t/h	2030 t/h	2242 t/h	2133 t/h	2004 t/h	1916 t/h
	54 sec	0.90 min	2189 t/h	2091 t/h	1973 t/h	1893 t/h	2066 t/h	1974 t/h	1862 t/h	1786 t/h
	66 sec	1.10 min	2030 t/h	1945 t/h	1843 t/h	1772 t/h	1916 t/h	1836 t/h	1739 t/h	1673 t/h
65%	30 sec	0.50 min	2410 t/h	2283 t/h	2133 t/h	2032 t/h	2275 t/h	2155 t/h	2013 t/h	1918 t/h
	42 sec	0.70 min	2205 t/h	2098 t/h	1971 t/h	1885 t/h	2081 t/h	1980 t/h	1860 t/h	1779 t/h
	54 sec	0.90 min	2032 t/h	1941 t/h	1831 t/h	1757 t/h	1918 t/h	1832 t/h	1728 t/h	1658 t/h
	66 sec	1.10 min	1885 t/h	1806 t/h	1710 t/h	1645 t/h	1779 t/h	1704 t/h	1614 t/h	1553 t/h

MODEL: 6030 with Face Shovel attachment ● Bucket Size: 21.6 yd³ ● Number of Cycles: 5 ● Material Density: 3030 lb/yd³

Overall Mine Efficiency	Bucket Fill Factor		100%				95%			
	Load per Bucket		32.7 tons				31.1 tons			
	Load per Truck		164 tons				155 tons			
	Cycle Time		24 sec	26 sec	28 sec	30 sec	24 sec	26 sec	28 sec	30 sec
	Truck Exchange		0.40 min	0.43 min	0.47 min	0.50 min	0.40 min	0.43 min	0.47 min	0.50 min
83% = maximum 50 min/hour	30 sec	0.50 min	3792 tons/h	3591 tons/h	3355 tons/h	3197 tons/h	3600 tons/h	3410 tons/h	3185 tons/h	3035 tons/h
	42 sec	0.70 min	3469 tons/h	3300 tons/h	3100 tons/h	2964 tons/h	3294 tons/h	3134 tons/h	2943 tons/h	2815 tons/h
	54 sec	0.90 min	3197 tons/h	3053 tons/h	2881 tons/h	2763 tons/h	3035 tons/h	2899 tons/h	2735 tons/h	2624 tons/h
	66 sec	1.10 min	2964 tons/h	2840 tons/h	2690 tons/h	2588 tons/h	2815 tons/h	2697 tons/h	2555 tons/h	2457 tons/h
80%	30 sec	0.50 min	3641 tons/h	3448 tons/h	3221 tons/h	3070 tons/h	3457 tons/h	3274 tons/h	3059 tons/h	2915 tons/h
	42 sec	0.70 min	3331 tons/h	3169 tons/h	2976 tons/h	2846 tons/h	3163 tons/h	3009 tons/h	2826 tons/h	2703 tons/h
	54 sec	0.90 min	3070 tons/h	2932 tons/h	2766 tons/h	2653 tons/h	2915 tons/h	2784 tons/h	2626 tons/h	2519 tons/h
	66 sec	1.10 min	2846 tons/h	2727 tons/h	2583 tons/h	2485 tons/h	2703 tons/h	2590 tons/h	2453 tons/h	2360 tons/h
75%	30 sec	0.50 min	3412 tons/h	3232 tons/h	3019 tons/h	2877 tons/h	3240 tons/h	3069 tons/h	2867 tons/h	2732 tons/h
	42 sec	0.70 min	3122 tons/h	2970 tons/h	2790 tons/h	2668 tons/h	2964 tons/h	2820 tons/h	2649 tons/h	2533 tons/h
	54 sec	0.90 min	2877 tons/h	2748 tons/h	2592 tons/h	2487 tons/h	2732 tons/h	2609 tons/h	2462 tons/h	2361 tons/h
	66 sec	1.10 min	2668 tons/h	2556 tons/h	2421 tons/h	2329 tons/h	2533 tons/h	2427 tons/h	2299 tons/h	2211 tons/h
70%	30 sec	0.50 min	3184 tons/h	3016 tons/h	2817 tons/h	2685 tons/h	3023 tons/h	2863 tons/h	2675 tons/h	2549 tons/h
	42 sec	0.70 min	2913 tons/h	2772 tons/h	2603 tons/h	2489 tons/h	2766 tons/h	2632 tons/h	2471 tons/h	2364 tons/h
	54 sec	0.90 min	2685 tons/h	2564 tons/h	2419 tons/h	2321 tons/h	2549 tons/h	2434 tons/h	2297 tons/h	2203 tons/h
	66 sec	1.10 min	2489 tons/h	2385 tons/h	2259 tons/h	2173 tons/h	2364 tons/h	2265 tons/h	2145 tons/h	2063 tons/h
65%	30 sec	0.50 min	2956 tons/h	2799 tons/h	2615 tons/h	2492 tons/h	2806 tons/h	2658 tons/h	2483 tons/h	2366 tons/h
	42 sec	0.70 min	2704 tons/h	2573 tons/h	2416 tons/h	2311 tons/h	2568 tons/h	2443 tons/h	2294 tons/h	2194 tons/h
	54 sec	0.90 min	2492 tons/h	2380 tons/h	2245 tons/h	2154 tons/h	2366 tons/h	2260 tons/h	2132 tons/h	2045 tons/h
	66 sec	1.10 min	2311 tons/h	2214 tons/h	2097 tons/h	2017 tons/h	2194 tons/h	2102 tons/h	1991 tons/h	1915 tons/h

Overall Mine Efficiency	Bucket Fill Factor		90%				85%			
	Load per Bucket		29.4 tons				27.8 tons			
	Load per Truck		147 tons				139 tons			
	Cycle Time		24 sec	26 sec	28 sec	30 sec	24 sec	26 sec	28 sec	30 sec
	Truck Exchange		0.40 min	0.43 min	0.47 min	0.50 min	0.40 min	0.43 min	0.47 min	0.50 min
83% = maximum 50 min/hour	30 sec	0.50 min	3409 tons/h	3228 tons/h	3016 tons/h	2874 tons/h	3217 tons/h	3047 tons/h	2846 tons/h	2712 tons/h
	42 sec	0.70 min	3119 tons/h	2967 tons/h	2786 tons/h	2665 tons/h	2943 tons/h	2800 tons/h	2630 tons/h	2515 tons/h
	54 sec	0.90 min	2874 tons/h	2745 tons/h	2590 tons/h	2484 tons/h	2712 tons/h	2591 tons/h	2444 tons/h	2345 tons/h
	66 sec	1.10 min	2665 tons/h	2553 tons/h	2419 tons/h	2327 tons/h	2515 tons/h	2410 tons/h	2283 tons/h	2196 tons/h
80%	30 sec	0.50 min	3273 tons/h	3100 tons/h	2896 tons/h	2760 tons/h	3089 tons/h	2926 tons/h	2733 tons/h	2605 tons/h
	42 sec	0.70 min	2995 tons/h	2849 tons/h	2676 tons/h	2559 tons/h	2826 tons/h	2689 tons/h	2525 tons/h	2415 tons/h
	54 sec	0.90 min	2760 tons/h	2636 tons/h	2487 tons/h	2385 tons/h	2605 tons/h	2488 tons/h	2347 tons/h	2251 tons/h
	66 sec	1.10 min	2559 tons/h	2452 tons/h	2322 tons/h	2234 tons/h	2415 tons/h	2314 tons/h	2192 tons/h	2109 tons/h
75%	30 sec	0.50 min	3068 tons/h	2906 tons/h	2714 tons/h	2587 tons/h	2895 tons/h	2742 tons/h	2562 tons/h	2441 tons/h
	42 sec	0.70 min	2807 tons/h	2670 tons/h	2508 tons/h	2398 tons/h	2649 tons/h	2520 tons/h	2367 tons/h	2264 tons/h
	54 sec	0.90 min	2587 tons/h	2470 tons/h	2331 tons/h	2236 tons/h	2441 tons/h	2331 tons/h	2200 tons/h	2110 tons/h
	66 sec	1.10 min	2398 tons/h	2298 tons/h	2177 tons/h	2094 tons/h	2264 tons/h	2169 tons/h	2054 tons/h	1976 tons/h
70%	30 sec	0.50 min	2862 tons/h	2711 tons/h	2533 tons/h	2413 tons/h	2702 tons/h	2559 tons/h	2390 tons/h	2278 tons/h
	42 sec	0.70 min	2619 tons/h	2492 tons/h	2340 tons/h	2238 tons/h	2472 tons/h	2352 tons/h	2209 tons/h	2112 tons/h
	54 sec	0.90 min	2413 tons/h	2305 tons/h	2175 tons/h	2086 tons/h	2278 tons/h	2175 tons/h	2052 tons/h	1969 tons/h
	66 sec	1.10 min	2238 tons/h	2144 tons/h	2031 tons/h	1954 tons/h	2112 tons/h	2024 tons/h	1917 tons/h	1844 tons/h
65%	30 sec	0.50 min	2657 tons/h	2517 tons/h	2351 tons/h	2240 tons/h	2508 tons/h	2375 tons/h	2219 tons/h	2114 tons/h
	42 sec	0.70 min	2431 tons/h	2313 tons/h	2172 tons/h	2077 tons/h	2294 tons/h	2183 tons/h	2050 tons/h	1961 tons/h
	54 sec	0.90 min	2240 tons/h	2140 tons/h	2019 tons/h	1937 tons/h	2114 tons/h	2019 tons/h	1905 tons/h	1828 tons/h
	66 sec	1.10 min	2077 tons/h	1990 tons/h	1885 tons/h	1814 tons/h	1961 tons/h	1879 tons/h	1779 tons/h	1712 tons/h

MODEL: 6040 ● Bucket Size: 22.0 m³ ● Number of Cycles: 4 ● Material Density: 1.80 t/m³

	Bucket Fill Factor		100%				95%			
	Load per Bucket		39.6 t				37.6 t			
	Load per Truck		158 t				150 t			
Overall Mine Efficiency	Cycle Time		24 sec	26 sec	28 sec	30 sec	24 sec	26 sec	28 sec	30 sec
	Truck Exchange		0.40 min	0.43 min	0.47 min	0.50 min	0.40 min	0.43 min	0.47 min	0.50 min
83% = maximum 50 min/hour	30 sec	0.50 min	4508 t/h	4287 t/h	4025 t/h	3848 t/h	4280 t/h	4071 t/h	3821 t/h	3654 t/h
	42 sec	0.70 min	4045 t/h	3867 t/h	3652 t/h	3506 t/h	3841 t/h	3672 t/h	3468 t/h	3329 t/h
	54 sec	0.90 min	3669 t/h	3522 t/h	3343 t/h	3220 t/h	3484 t/h	3344 t/h	3174 t/h	3057 t/h
	66 sec	1.10 min	3357 t/h	3233 t/h	3081 t/h	2977 t/h	3187 t/h	3070 t/h	2926 t/h	2826 t/h
80%	30 sec	0.50 min	4328 t/h	4117 t/h	3865 t/h	3695 t/h	4110 t/h	3909 t/h	3669 t/h	3508 t/h
	42 sec	0.70 min	3884 t/h	3713 t/h	3507 t/h	3367 t/h	3688 t/h	3526 t/h	3330 t/h	3197 t/h
	54 sec	0.90 min	3523 t/h	3382 t/h	3210 t/h	3092 t/h	3345 t/h	3211 t/h	3048 t/h	2936 t/h
	66 sec	1.10 min	3223 t/h	3104 t/h	2959 t/h	2858 t/h	3060 t/h	2948 t/h	2809 t/h	2714 t/h
75%	30 sec	0.50 min	4057 t/h	3858 t/h	3622 t/h	3463 t/h	3852 t/h	3664 t/h	3439 t/h	3288 t/h
	42 sec	0.70 min	3641 t/h	3480 t/h	3287 t/h	3155 t/h	3457 t/h	3304 t/h	3121 t/h	2996 t/h
	54 sec	0.90 min	3302 t/h	3169 t/h	3008 t/h	2898 t/h	3135 t/h	3009 t/h	2856 t/h	2751 t/h
	66 sec	1.10 min	3021 t/h	2910 t/h	2773 t/h	2679 t/h	2868 t/h	2763 t/h	2633 t/h	2544 t/h
70%	30 sec	0.50 min	3785 t/h	3600 t/h	3380 t/h	3231 t/h	3594 t/h	3418 t/h	3209 t/h	3068 t/h
	42 sec	0.70 min	3397 t/h	3247 t/h	3067 t/h	2944 t/h	3226 t/h	3083 t/h	2912 t/h	2795 t/h
	54 sec	0.90 min	3081 t/h	2957 t/h	2807 t/h	2704 t/h	2925 t/h	2808 t/h	2665 t/h	2567 t/h
	66 sec	1.10 min	2819 t/h	2715 t/h	2588 t/h	2500 t/h	2676 t/h	2578 t/h	2457 t/h	2373 t/h
65%	30 sec	0.50 min	3514 t/h	3342 t/h	3137 t/h	3000 t/h	3336 t/h	3173 t/h	2979 t/h	2848 t/h
	42 sec	0.70 min	3153 t/h	3014 t/h	2847 t/h	2733 t/h	2994 t/h	2862 t/h	2703 t/h	2595 t/h
	54 sec	0.90 min	2860 t/h	2745 t/h	2606 t/h	2510 t/h	2716 t/h	2606 t/h	2474 t/h	2383 t/h
	66 sec	1.10 min	2617 t/h	2520 t/h	2402 t/h	2320 t/h	2484 t/h	2393 t/h	2281 t/h	2203 t/h

	Bucket Fill Factor		90%				85%			
	Load per Bucket		35.6 t				33.7 t			
	Load per Truck		142 t				135 t			
Overall Mine Efficiency	Cycle Time		24 sec	26 sec	28 sec	30 sec	24 sec	26 sec	28 sec	30 sec
	Truck Exchange		0.40 min	0.43 min	0.47 min	0.50 min	0.40 min	0.43 min	0.47 min	0.50 min
83% = maximum 50 min/hour	30 sec	0.50 min	4052 t/h	3854 t/h	3618 t/h	3459 t/h	3836 t/h	3648 t/h	3425 t/h	3275 t/h
	42 sec	0.70 min	3637 t/h	3476 t/h	3283 t/h	3152 t/h	3443 t/h	3291 t/h	3108 t/h	2984 t/h
	54 sec	0.90 min	3298 t/h	3166 t/h	3005 t/h	2894 t/h	3122 t/h	2997 t/h	2845 t/h	2740 t/h
	66 sec	1.10 min	3018 t/h	2906 t/h	2770 t/h	2676 t/h	2857 t/h	2751 t/h	2622 t/h	2533 t/h
80%	30 sec	0.50 min	3891 t/h	3701 t/h	3474 t/h	3322 t/h	3684 t/h	3503 t/h	3289 t/h	3144 t/h
	42 sec	0.70 min	3492 t/h	3338 t/h	3153 t/h	3026 t/h	3306 t/h	3160 t/h	2984 t/h	2865 t/h
	54 sec	0.90 min	3167 t/h	3040 t/h	2885 t/h	2779 t/h	2998 t/h	2878 t/h	2731 t/h	2631 t/h
	66 sec	1.10 min	2898 t/h	2791 t/h	2660 t/h	2570 t/h	2743 t/h	2642 t/h	2518 t/h	2433 t/h
75%	30 sec	0.50 min	3647 t/h	3469 t/h	3256 t/h	3113 t/h	3452 t/h	3284 t/h	3083 t/h	2947 t/h
	42 sec	0.70 min	3273 t/h	3129 t/h	2955 t/h	2837 t/h	3098 t/h	2962 t/h	2797 t/h	2685 t/h
	54 sec	0.90 min	2969 t/h	2849 t/h	2704 t/h	2605 t/h	2810 t/h	2697 t/h	2560 t/h	2466 t/h
	66 sec	1.10 min	2716 t/h	2616 t/h	2493 t/h	2408 t/h	2571 t/h	2476 t/h	2360 t/h	2280 t/h
70%	30 sec	0.50 min	3403 t/h	3237 t/h	3038 t/h	2905 t/h	3221 t/h	3064 t/h	2876 t/h	2750 t/h
	42 sec	0.70 min	3054 t/h	2919 t/h	2757 t/h	2647 t/h	2891 t/h	2763 t/h	2610 t/h	2505 t/h
	54 sec	0.90 min	2770 t/h	2659 t/h	2523 t/h	2431 t/h	2622 t/h	2517 t/h	2389 t/h	2301 t/h
	66 sec	1.10 min	2534 t/h	2441 t/h	2326 t/h	2247 t/h	2399 t/h	2310 t/h	2202 t/h	2127 t/h
65%	30 sec	0.50 min	3159 t/h	3004 t/h	2820 t/h	2697 t/h	2990 t/h	2844 t/h	2670 t/h	2553 t/h
	42 sec	0.70 min	2835 t/h	2710 t/h	2559 t/h	2457 t/h	2684 t/h	2565 t/h	2423 t/h	2326 t/h
	54 sec	0.90 min	2571 t/h	2468 t/h	2342 t/h	2256 t/h	2434 t/h	2336 t/h	2217 t/h	2136 t/h
	66 sec	1.10 min	2352 t/h	2266 t/h	2159 t/h	2086 t/h	2227 t/h	2145 t/h	2044 t/h	1975 t/h

MODEL: 6040 ● Bucket Size: 28.8 yd³ ● Number of Cycles: 4 ● Material Density: 3030 lb/yd³

Overall Mine Efficiency	Truck Exchange / Cycle Time		24 sec	26 sec	28 sec	30 sec	24 sec	26 sec	28 sec	30 sec
	Bucket Fill Factor		**100%**				**95%**			
	Load per Bucket		43.7 tons				41.4 tons			
	Load per Truck		175 tons				166 tons			
	Truck Exchange		**0.40 min**	**0.43 min**	**0.47 min**	**0.50 min**	**0.40 min**	**0.43 min**	**0.47 min**	**0.50 min**
83% = maximum 50 min/hour	30 sec	**0.50 min**	4969 tons/h	4726 tons/h	4436 tons/h	4242 tons/h	4718 tons/h	4487 tons/h	4212 tons/h	4027 tons/h
	42 sec	**0.70 min**	4459 tons/h	4262 tons/h	4026 tons/h	3865 tons/h	4234 tons/h	4047 tons/h	3822 tons/h	3669 tons/h
	54 sec	**0.90 min**	4044 tons/h	3882 tons/h	3684 tons/h	3549 tons/h	3840 tons/h	3686 tons/h	3498 tons/h	3370 tons/h
	66 sec	**1.10 min**	3700 tons/h	3564 tons/h	3397 tons/h	3281 tons/h	3513 tons/h	3384 tons/h	3225 tons/h	3116 tons/h
80%	30 sec	**0.50 min**	4771 tons/h	4538 tons/h	4260 tons/h	4073 tons/h	4530 tons/h	4309 tons/h	4045 tons/h	3867 tons/h
	42 sec	**0.70 min**	4282 tons/h	4093 tons/h	3866 tons/h	3711 tons/h	4066 tons/h	3886 tons/h	3670 tons/h	3524 tons/h
	54 sec	**0.90 min**	3884 tons/h	3728 tons/h	3538 tons/h	3408 tons/h	3687 tons/h	3539 tons/h	3359 tons/h	3236 tons/h
	66 sec	**1.10 min**	3553 tons/h	3422 tons/h	3262 tons/h	3151 tons/h	3374 tons/h	3249 tons/h	3097 tons/h	2992 tons/h
75%	30 sec	**0.50 min**	4472 tons/h	4253 tons/h	3993 tons/h	3817 tons/h	4246 tons/h	4038 tons/h	3791 tons/h	3625 tons/h
	42 sec	**0.70 min**	4013 tons/h	3836 tons/h	3623 tons/h	3478 tons/h	3811 tons/h	3642 tons/h	3440 tons/h	3302 tons/h
	54 sec	**0.90 min**	3640 tons/h	3494 tons/h	3316 tons/h	3194 tons/h	3456 tons/h	3317 tons/h	3149 tons/h	3033 tons/h
	66 sec	**1.10 min**	3330 tons/h	3207 tons/h	3057 tons/h	2953 tons/h	3162 tons/h	3045 tons/h	2903 tons/h	2804 tons/h
70%	30 sec	**0.50 min**	4173 tons/h	3968 tons/h	3726 tons/h	3562 tons/h	3962 tons/h	3768 tons/h	3537 tons/h	3382 tons/h
	42 sec	**0.70 min**	3745 tons/h	3579 tons/h	3381 tons/h	3245 tons/h	3556 tons/h	3399 tons/h	3210 tons/h	3081 tons/h
	54 sec	**0.90 min**	3396 tons/h	3260 tons/h	3094 tons/h	2980 tons/h	3225 tons/h	3095 tons/h	2938 tons/h	2830 tons/h
	66 sec	**1.10 min**	3107 tons/h	2993 tons/h	2852 tons/h	2755 tons/h	2950 tons/h	2841 tons/h	2708 tons/h	2616 tons/h
65%	30 sec	**0.50 min**	3873 tons/h	3684 tons/h	3458 tons/h	3306 tons/h	3678 tons/h	3498 tons/h	3284 tons/h	3139 tons/h
	42 sec	**0.70 min**	3476 tons/h	3323 tons/h	3138 tons/h	3013 tons/h	3300 tons/h	3155 tons/h	2980 tons/h	2860 tons/h
	54 sec	**0.90 min**	3153 tons/h	3026 tons/h	2872 tons/h	2767 tons/h	2993 tons/h	2873 tons/h	2727 tons/h	2627 tons/h
	66 sec	**1.10 min**	2884 tons/h	2778 tons/h	2648 tons/h	2558 tons/h	2739 tons/h	2638 tons/h	2514 tons/h	2429 tons/h

Overall Mine Efficiency	Truck Exchange / Cycle Time		24 sec	26 sec	28 sec	30 sec	24 sec	26 sec	28 sec	30 sec
	Bucket Fill Factor		**90%**				**85%**			
	Load per Bucket		39.2 tons				37.1 tons			
	Load per Truck		157 tons				149 tons			
	Truck Exchange		**0.40 min**	**0.43 min**	**0.47 min**	**0.50 min**	**0.40 min**	**0.43 min**	**0.47 min**	**0.50 min**
83% = maximum 50 min/hour	30 sec	**0.50 min**	4467 tons/h	4248 tons/h	3988 tons/h	3813 tons/h	4228 tons/h	4022 tons/h	3775 tons/h	3610 tons/h
	42 sec	**0.70 min**	4009 tons/h	3832 tons/h	3619 tons/h	3474 tons/h	3795 tons/h	3627 tons/h	3426 tons/h	3289 tons/h
	54 sec	**0.90 min**	3636 tons/h	3490 tons/h	3312 tons/h	3191 tons/h	3442 tons/h	3304 tons/h	3136 tons/h	3020 tons/h
	66 sec	**1.10 min**	3326 tons/h	3204 tons/h	3054 tons/h	2950 tons/h	3149 tons/h	3033 tons/h	2891 tons/h	2792 tons/h
80%	30 sec	**0.50 min**	4289 tons/h	4079 tons/h	3830 tons/h	3662 tons/h	4060 tons/h	3862 tons/h	3625 tons/h	3466 tons/h
	42 sec	**0.70 min**	3849 tons/h	3680 tons/h	3475 tons/h	3336 tons/h	3644 tons/h	3483 tons/h	3290 tons/h	3158 tons/h
	54 sec	**0.90 min**	3491 tons/h	3351 tons/h	3181 tons/h	3064 tons/h	3305 tons/h	3172 tons/h	3011 tons/h	2900 tons/h
	66 sec	**1.10 min**	3194 tons/h	3076 tons/h	2932 tons/h	2833 tons/h	3024 tons/h	2912 tons/h	2776 tons/h	2681 tons/h
75%	30 sec	**0.50 min**	4020 tons/h	3824 tons/h	3589 tons/h	3432 tons/h	3806 tons/h	3619 tons/h	3398 tons/h	3249 tons/h
	42 sec	**0.70 min**	3608 tons/h	3449 tons/h	3257 tons/h	3127 tons/h	3415 tons/h	3265 tons/h	3083 tons/h	2960 tons/h
	54 sec	**0.90 min**	3272 tons/h	3141 tons/h	2981 tons/h	2872 tons/h	3098 tons/h	2973 tons/h	2822 tons/h	2718 tons/h
	66 sec	**1.10 min**	2994 tons/h	2883 tons/h	2748 tons/h	2655 tons/h	2834 tons/h	2729 tons/h	2602 tons/h	2513 tons/h
70%	30 sec	**0.50 min**	3751 tons/h	3568 tons/h	3349 tons/h	3202 tons/h	3551 tons/h	3377 tons/h	3170 tons/h	3031 tons/h
	42 sec	**0.70 min**	3366 tons/h	3218 tons/h	3039 tons/h	2918 tons/h	3187 tons/h	3046 tons/h	2877 tons/h	2762 tons/h
	54 sec	**0.90 min**	3053 tons/h	2931 tons/h	2782 tons/h	2679 tons/h	2890 tons/h	2774 tons/h	2633 tons/h	2536 tons/h
	66 sec	**1.10 min**	2793 tons/h	2690 tons/h	2564 tons/h	2477 tons/h	2644 tons/h	2547 tons/h	2427 tons/h	2345 tons/h
65%	30 sec	**0.50 min**	3482 tons/h	3312 tons/h	3109 tons/h	2972 tons/h	3296 tons/h	3135 tons/h	2943 tons/h	2814 tons/h
	42 sec	**0.70 min**	3125 tons/h	2987 tons/h	2821 tons/h	2708 tons/h	2958 tons/h	2828 tons/h	2671 tons/h	2564 tons/h
	54 sec	**0.90 min**	2834 tons/h	2720 tons/h	2582 tons/h	2487 tons/h	2683 tons/h	2575 tons/h	2444 tons/h	2354 tons/h
	66 sec	**1.10 min**	2593 tons/h	2497 tons/h	2380 tons/h	2299 tons/h	2455 tons/h	2364 tons/h	2253 tons/h	2177 tons/h

MODEL: 6060 ● Bucket Size: 34.0 m³ ● Number of Cycles: 4 ● Material Density: 1.80 t/m³

	Bucket Fill Factor		100%				95%			
	Load per Bucket		61.2 t				58.1 t			
	Load per Truck		245 t				232 t			
Overall Mine Efficiency	Cycle Time		26 sec	28 sec	30 sec	32 sec	26 sec	28 sec	30 sec	32 sec
	Truck Exchange		0.43 min	0.47 min	0.50 min	0.53 min	0.43 min	0.47 min	0.50 min	0.53 min
83% = maximum 50 min/hour	30 sec	0.50 min	6626 t/h	6220 t/h	5947 t/h	5697 t/h	6290 t/h	5905 t/h	5646 t/h	5408 t/h
	42 sec	0.70 min	5976 t/h	5644 t/h	5418 t/h	5210 t/h	5673 t/h	5358 t/h	5144 t/h	4946 t/h
	54 sec	0.90 min	5442 t/h	5166 t/h	4976 t/h	4800 t/h	5167 t/h	4904 t/h	4724 t/h	4557 t/h
	66 sec	1.10 min	4996 t/h	4762 t/h	4600 t/h	4449 t/h	4743 t/h	4521 t/h	4367 t/h	4224 t/h
80%	30 sec	0.50 min	6362 t/h	5973 t/h	5710 t/h	5470 t/h	6040 t/h	5670 t/h	5421 t/h	5193 t/h
	42 sec	0.70 min	5738 t/h	5420 t/h	5203 t/h	5003 t/h	5448 t/h	5145 t/h	4939 t/h	4749 t/h
	54 sec	0.90 min	5226 t/h	4960 t/h	4778 t/h	4609 t/h	4961 t/h	4709 t/h	4536 t/h	4375 t/h
	66 sec	1.10 min	4798 t/h	4573 t/h	4417 t/h	4272 t/h	4555 t/h	4341 t/h	4194 t/h	4056 t/h
75%	30 sec	0.50 min	5963 t/h	5598 t/h	5352 t/h	5127 t/h	5661 t/h	5314 t/h	5081 t/h	4867 t/h
	42 sec	0.70 min	5378 t/h	5080 t/h	4876 t/h	4689 t/h	5106 t/h	4822 t/h	4629 t/h	4451 t/h
	54 sec	0.90 min	4898 t/h	4649 t/h	4478 t/h	4320 t/h	4650 t/h	4414 t/h	4251 t/h	4101 t/h
	66 sec	1.10 min	4497 t/h	4286 t/h	4140 t/h	4004 t/h	4269 t/h	4069 t/h	3931 t/h	3802 t/h
70%	30 sec	0.50 min	5564 t/h	5223 t/h	4994 t/h	4784 t/h	5282 t/h	4959 t/h	4741 t/h	4542 t/h
	42 sec	0.70 min	5018 t/h	4740 t/h	4550 t/h	4375 t/h	4764 t/h	4500 t/h	4320 t/h	4153 t/h
	54 sec	0.90 min	4570 t/h	4338 t/h	4179 t/h	4031 t/h	4339 t/h	4118 t/h	3967 t/h	3826 t/h
	66 sec	1.10 min	4196 t/h	3999 t/h	3863 t/h	3736 t/h	3983 t/h	3796 t/h	3668 t/h	3547 t/h
65%	30 sec	0.50 min	5165 t/h	4849 t/h	4636 t/h	4441 t/h	4903 t/h	4603 t/h	4401 t/h	4216 t/h
	42 sec	0.70 min	4658 t/h	4400 t/h	4224 t/h	4061 t/h	4422 t/h	4177 t/h	4010 t/h	3855 t/h
	54 sec	0.90 min	4242 t/h	4027 t/h	3879 t/h	3741 t/h	4028 t/h	3823 t/h	3682 t/h	3552 t/h
	66 sec	1.10 min	3895 t/h	3712 t/h	3586 t/h	3468 t/h	3697 t/h	3524 t/h	3404 t/h	3293 t/h

	Bucket Fill Factor		90%				85%			
	Load per Bucket		55.1 t				52.0 t			
	Load per Truck		220 t				208 t			
Overall Mine Efficiency	Cycle Time		26 sec	28 sec	30 sec	32 sec	26 sec	28 sec	30 sec	32 sec
	Truck Exchange		0.43 min	0.47 min	0.50 min	0.53 min	0.43 min	0.47 min	0.50 min	0.53 min
83% = maximum 50 min/hour	30 sec	0.50 min	5965 t/h	5600 t/h	5354 t/h	5129 t/h	5630 t/h	5285 t/h	5053 t/h	4840 t/h
	42 sec	0.70 min	5380 t/h	5081 t/h	4878 t/h	4691 t/h	5078 t/h	4796 t/h	4604 t/h	4427 t/h
	54 sec	0.90 min	4900 t/h	4651 t/h	4480 t/h	4321 t/h	4624 t/h	4389 t/h	4228 t/h	4078 t/h
	66 sec	1.10 min	4498 t/h	4287 t/h	4142 t/h	4006 t/h	4245 t/h	4046 t/h	3909 t/h	3780 t/h
80%	30 sec	0.50 min	5728 t/h	5377 t/h	5141 t/h	4925 t/h	5406 t/h	5075 t/h	4852 t/h	4648 t/h
	42 sec	0.70 min	5166 t/h	4879 t/h	4684 t/h	4504 t/h	4876 t/h	4605 t/h	4421 t/h	4251 t/h
	54 sec	0.90 min	4705 t/h	4466 t/h	4302 t/h	4149 t/h	4440 t/h	4215 t/h	4060 t/h	3916 t/h
	66 sec	1.10 min	4319 t/h	4117 t/h	3977 t/h	3847 t/h	4076 t/h	3885 t/h	3753 t/h	3630 t/h
75%	30 sec	0.50 min	5369 t/h	5040 t/h	4819 t/h	4616 t/h	5067 t/h	4756 t/h	4548 t/h	4356 t/h
	42 sec	0.70 min	4842 t/h	4573 t/h	4390 t/h	4222 t/h	4570 t/h	4316 t/h	4143 t/h	3984 t/h
	54 sec	0.90 min	4410 t/h	4186 t/h	4032 t/h	3889 t/h	4162 t/h	3950 t/h	3805 t/h	3670 t/h
	66 sec	1.10 min	4048 t/h	3859 t/h	3728 t/h	3605 t/h	3821 t/h	3642 t/h	3518 t/h	3402 t/h
70%	30 sec	0.50 min	5009 t/h	4703 t/h	4496 t/h	4307 t/h	4727 t/h	4438 t/h	4243 t/h	4065 t/h
	42 sec	0.70 min	4518 t/h	4267 t/h	4097 t/h	3939 t/h	4264 t/h	4027 t/h	3866 t/h	3717 t/h
	54 sec	0.90 min	4115 t/h	3906 t/h	3762 t/h	3629 t/h	3883 t/h	3686 t/h	3550 t/h	3425 t/h
	66 sec	1.10 min	3778 t/h	3600 t/h	3478 t/h	3364 t/h	3565 t/h	3398 t/h	3282 t/h	3175 t/h
65%	30 sec	0.50 min	4650 t/h	4365 t/h	4174 t/h	3998 t/h	4388 t/h	4120 t/h	3939 t/h	3773 t/h
	42 sec	0.70 min	4194 t/h	3961 t/h	3803 t/h	3656 t/h	3958 t/h	3738 t/h	3589 t/h	3451 t/h
	54 sec	0.90 min	3820 t/h	3625 t/h	3492 t/h	3368 t/h	3605 t/h	3421 t/h	3296 t/h	3179 t/h
	66 sec	1.10 min	3507 t/h	3342 t/h	3229 t/h	3123 t/h	3309 t/h	3154 t/h	3047 t/h	2947 t/h